INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH

WILEY

INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH

Intl. Trans. in Op. Res. 33 (2026) 719–774
DOI: 10.1111/itor.70054

Pitfalls, benefits, and comparative analysis of artificial intelligence ChatBots in the systematic review process

Cinzia Daraio*, Simone Di Leo and Federico Ferazzoli

Department of Computer, Control and Management Engineering Antonio Ruberti (DIAG), Sapienza University of Rome, Via Ariosto 25, Rome 00185, Italy
E-mail: daraio@diag.uniroma1.it [Daraio]; dileo@diag.uniroma1.it [Di Leo]; ferazzoli.1918794@studenti.uniro [Ferazzoli]

Received 28 August 2024; received in revised form 30 March 2025; accepted 2 May 2025

Abstract

The transformative potential of artificial intelligence (AI) ChatBots, leveraging natural language processing for information retrieval and knowledge synthesis, has garnered attention across diverse fields, including research. Recognizing AI's importance, researchers and policymakers are actively engaged in its development. However, a notable gap persists in understanding the efficacy of various AI-ChatBots specifically within the rigorous process of conducting systematic literature reviews (SLRs). To address this, this study aims to provide a comparative analysis of different AI-ChatBots against traditional manual approaches in executing SLRs. Employing a case study methodology focused on the *International Transactions in Operational Research* journal, this research evaluates the benefits and pitfalls of integrating AI across various stages of the SLR process. The findings of this study will offer novel empirical evidence regarding the application of AI-ChatBots in SLR processes, thereby informing both researchers and policymakers on the practical implications and future directions of AI integration in academic research.

Keywords: systematic review; artificial intelligence; ChatBots; *International Transactions in Operational Research*

1. Introduction and aims

The rapid advancement of artificial intelligence (AI) technologies has ushered in a new era of possibilities for scientific research. Among these innovations, ChatBots powered by AI, mainly powered by large language models (LLMs), a subfield of the AI field, henceforth referred to as AI-ChatBots, have gathered significant attention for their potential to revolutionize various aspects of the research process. This capability has sparked wide interest not only among researchers but also in various fields where research results are applied. The versatility of these tools makes them

*Corresponding author.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.
This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

potentially suitable for a wide range of scientific research applications. The evolution of conversational and generative AI (GenAI) technologies like ChatGPT (OpenAI, 2022) and their impact on education and research has been recently analyzed by Akpan et al. (2025). The growing importance of AI in research has not gone unnoticed by policymakers. In Italy, the Strategic Program for AI 2022–24 (Governo Italiano, 2021) outlines goals and strategies for both the creation of new GenAI tools and training for people with expertise in the field. At the international level, the European Union has issued the first text on AI regulation (European Parliament, 2023), defining areas of unacceptable risk AI creation and high-risk AI systems. The Vatican State has also intervened with a call on AI ethics (Vatican State, 2020), emphasizing the need for responsible development and use of these technologies. Hosseini et al. (2023) analyze the ethical implications associated with the use of AI in the writing of scientific papers.

As the volume of scientific literature continues to grow exponentially, researchers are faced with ever-increasing challenges to efficiently summarize and analyze existing knowledge. This challenge is particularly felt in systematic reviews (Ioannidis et al., 2015). The laborious nature of systematic reviews has long been recognized as a limitation, potentially delaying the dissemination of critical syntheses of research findings (Borah et al., 2017). As defined by Avenali et al. (2023), systematic literature reviews (SLRs) consist of a rigorous and precise sequence of steps designed to comprehensively understand and analyze a specific research topic. These steps typically include a planning phase in which researchers brainstorm to formulate the research question and plan the activities to be carried out for the review; literature search, screening and selection of studies; extraction and synthesis of the information and data needed to answer the research question; and drafting of the report in a manuscript. However, the process of conducting these reviews is often time-consuming and labor-intensive, requiring researchers to sift through thousands of potential articles, extract relevant information, and synthesize the results (Page et al., 2021). The increasing volume of scientific literature exacerbates these challenges. This growth not only increases the time needed to complete reviews but also raises concerns about the completeness and timeliness of reviews at the time of publication (Shojania et al., 2007). AI-ChatBots might present an interesting opportunity to improve this process while offering both interesting possibilities and significant challenges.

AI-ChatBots offer several potential benefits within the SLR process by analyzing textual data. These benefits extend beyond simply reading abstracts and can encompass various stages of the review including the screening and eligibility assessment, the data extraction, and the summary report phases.

Screening and eligibility assessment are critical steps in an SLR, where ChatBots can assist by analyzing abstracts, titles, keywords, and full texts to determine relevance to a research question. By processing large volumes of literature rapidly, AI models have the potential to accelerate what is often a time-consuming bottleneck in SLRs. Recent studies (Syriani et al., 2024) have demonstrated promising results in automating this process. Beyond screening, ChatBots can be employed for data extraction. By training the chatbot on a specific data extraction form, it can be instructed to identify and extract key information from the included studies, such as study design, outcome or even the article main topic. This automated data extraction can reduce manual effort and potentially minimize human error. While current chatbot capabilities are limited regarding direct synthesis of findings (e.g., meta-analysis), they can assist in organizing extracted data and identifying patterns or themes across studies. This could involve summarizing key findings or creating a report based on the previously extracted data.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

These tools offer the potential to significantly enhance efficiency and potentially reduce bias in certain stages of the review process. However, while AI tools can enhance efficiency and potentially reduce bias in certain review stages, they should be viewed as *assistive technologies* rather than replacements for human expertise. Human supervision remains essential, particularly in tasks such as screening, where a judgment is necessary (Syriani et al., 2024).

Several text analysis tools are on the market to help researchers in SLR, such as NVIVO (https://lumivero.com/products/nvivo) with good results (O'Neill et al., 2018). The main problem with these tools, however, is that they are not easy to use. Tools like NVIVO require training in qualitative data analysis methodologies. In contrast, AI-ChatBots are more user-friendly and accessible to researchers without extensive technical expertise.

However, realizing this potential will require careful consideration of ethical implications, rigorous evaluation of AI tools' performance, and thoughtful integration of AI capabilities with human expertise. The literature focuses mainly on ChatGPT, neither evaluating nor comparing how the different AI-ChatBots available behave in this area.

Despite the growing interest in AI applications for research, there is a significant gap in the literature regarding the specific potential and pitfalls of using these AI-ChatBots in conducting SLRs. While some studies have explored the use of specific ChatBot (like ChatGPT) in related topics, there is a lack of a comprehensive examination of how different AI-ChatBots can be applied in the various stages of an SLR and how the various AI-ChatBots available perform. Given the context outlined above, this study aims to address the following research questions:

1. What are the potential benefits and pitfalls of using AI-ChatBots in conducting or enhancing the systematic review process?
2. Which of the main AI-ChatBots available perform best in the SLR context?

To address these questions, we perform a comparative analysis between a manual review conducted by (Avenali et al., 2023) and a review supported by AI-ChatBots in each stage of the SLR process. This comparison regards a *simplified* SLR of the 30 most cited articles in the journal *International Transactions in Operational Research* (ITOR).

This research aims to make several significant contributions to the literature addressing the use of AI in the academic world. The contributions are:

- A *systematic comparison*. By systematically evaluating AI-ChatBots performances across the main stages of the SLR process and comparing them with a manual systematic review, this study provides a complete understanding of the tool's potential applications and limitations in this context. Also, the study provides an overview of the performance of the main free AI-ChatBots available to provide guidelines and suggestions on their *best* use in SLR.
- An *analysis of the potential benefits and pitfalls of AI*. The study explores the potential benefits of the use of AI tools in systematic reviews, considering how AI might be integrated to enhance rather than replace human expertise.
- Presents *new empirical evidence*. The research offers new empirical evidence on the use of AI tools in the review processes, helping to inform decision-making about when and how to incorporate AI technologies in the review process.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The article is organized as follows. Section 2 describes the background of AI, AI-ChatBots and their application in the academic research field. Section 3 describes the methodology and the data analyzed. Section 4 reports the results. Section 5 discusses the results obtained, Section 6 presents the threats to the validity of the findings, and Section 7 concludes the paper. Five appendices (A, B, C, D, and E) can be found at the end of the article for more details on the analyses presented.

2. Existing literature on AI and AI-ChatBots in academic research

In this section, after introducing what AI, LLMs, and AI-ChatBots are, the literature on the use of AI-ChatBots in academic research is presented.

2.1. What are AI and LLMs

AI is one of the most discussed topics in recent years as shown by the increasing number of scientific articles published on the topic.

A definition of AI is difficult to give, as the literature does not share a single explanation in the various disciplinary and application fields of this technology. The difficulty in defining AI is a consequence of the lack of a unified and clear concept of what human intelligence is. This is a long debate in academia that does not yet have an unambiguous answer (Schank, 1987) and would require rigorous philosophical investigation that is beyond our scope.

In this paper, we adopt the definition of AI presented by Sheikh et al. (2023), who defined AI in the field of computer science as follows: *AI stands for the imitation by computers of the intelligence inherent in humans*.

Building on this concept of AI imitating human intelligence, language models emerge as powerful tools for analyzing and interpreting textual data, mimicking human-like understanding of language. These linguistic models are probabilistic representations of natural language, designed to capture the intricacies and patterns of human communication. In this field, LLMs have emerged as a particularly advanced subset of AI technologies, pushing the boundaries of artificial linguistic capabilities (Radford et al., 2019). LMMs are a probability distribution over sequences of words. By analyzing a large corpus of text, the model can understand language in a mathematical sense. LLMs represent a significant evolution of traditional language models, distinguished by their unprecedented scale and complexity. The key to their enhanced performance lies in their training process: They are exposed to vast amounts of text during an intensive computational procedure that combines self-supervised and semi-supervised learning approaches. This extensive training allows LLMs to develop a deeper and context-aware understanding of language, further advancing the AI goal of imitating human intelligence in the domain of natural language processing. A significant breakthrough in the evolution from language models to LLMs came with the introduction of transformer architectures, proposed by Vaswani (2017). These architectures have had a profound impact on natural language processing and generative tasks, forming the basis for models like generative pre-trained transformer (GPT) (Han et al., 2021). The transformer architecture efficiently captures dependencies and relationships in sequential data, playing a vital role in the success of generative models, especially in natural language process applications.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The development of LLMs involves training on massive datasets, often comprising millions (or even billions) of parameters (Shen et al., 2023). One of the main problems in the LLM world is the quality of the data used. Like any statistical model, the quality of pre-training data is an important factor in the training of powerful LLMs, although this concept in the LLM domain has not been fully characterized (Lee et al., 2023). Poor quality of the data used can lead to biases, errors, and unwanted output (Raiaan et al., 2024).

This aspect, now more than ever, with the increasing spread of AI-generated content, may create problems in the future. Indeed, if LLMs are trained on datasets *dirtied* by AI data, they will collapse (Shumailov et al., 2024). More than 30 years ago, Schank (1991) criticized narrow and often superficial views of AI that focus on flashy demonstrations or simple rule-based systems. He advocated a deeper, more practical approach to AI that emphasized learning, scale, and representation of real-world knowledge. To progress, AI, Schank (1991) argued, must move from theoretical concepts to scalable, real-world applications that truly reflect intelligent behavior.

Despite this, when presented with a question or request, LLMs can generate relevant and meaningful answers by drawing on the learned linguistic models and representations. This capability makes them particularly useful in various applications, including the creation of meta-knowledge and systematic review. In recent years, there has also been growing attention on GenAI. GenAI, which encompasses LLMs, refers to AI technologies capable of creating new content such as text, images, music, speech, video, or code. What sets GenAI apart from traditional AI systems is its ability to create new, original content that is not explicitly programmed. This generative capability allows them to produce outputs similar in style, tone, or structure to the given prompt or instruction. The potential of GenAI and LLMs to amplify human capabilities in various domains is significant. However, it is crucial to approach their development and application responsibly, considering potential ethical implications (Kumar et al., 2023).

One of the main problems non-expert users had in the past years was the possibility of easy access to these tools. In this sense, the idea of *chat-based human-computer interaction* was the route taken by many companies. The use of chat-based tools (called Chatbots) was initially the idea of one of the most important pioneers in the field of computer science: Alan Turing in 1950 (Turing, 2009). As the name suggests, chatbots are software applications that simulate conversation or interaction with users, allowing them to have a human touch experience in conversation with these bots. There are different types of chatbots, from those based on patterns to those powered by AI or LLMs (Adamopoulou and Moussiades, 2020). The introduction of chat for access to LLMs revolutionized (and is still doing so) the world. This is because it has made the various LLMs accessible to many non-expert users. In this work, we focus the attention on the free version of the main AI-ChatBots available. The main reason behind the focus on the AI-ChatBots (and not on the models) is strictly related to the SLR process or in academic research in general. Not all researchers involved in SLR have the experience or expertise to access the various LLMs directly except through these *easy-to-use* interfaces. As pointed out by Bolaños et al. (2024), many tools exist to help researchers in SLR, but they are often lacking in terms of usability and user-friendliness, which limits their adoption by the broader research community. ChatBots, on the contrary, are easy to use and solve the problem related to the barrier of entry to these tools. The decision to instead analyze the free (and not the paid) versions of these AI-ChatBots concerns the accessibility that these tools have. The possibility of using paid chatbot is a particularly sensitive issue for researchers with budget constraints or researchers from developing countries (Alordiah, 2023). Moreover, of all the tools analyzed, the

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

free versions allow access to the most powerful versions of their underlying models in any case, with the main advantage given to paid users of avoiding technical problems such as the maximum number of words to be used or the waiting time between queries. Furthermore, the focus on these tools also relates to the main reason why users use chatbots (and one of the main problems with systematic reviews): increase productivity (Brandtzaeg and Følstad, 2017).

2.2. AI-ChatBots and its applications in academic research

The rapid advancement of AI lit considerable interest and debate in academics, particularly regarding its potential applications in research methodologies. One area where AI-ChatBots show promise is in enhancing the process of SLR, a foundation stone of evidence-based practice across various disciplines. As the volume of scientific literature continues to grow exponentially, researchers face increasing challenges in efficiently synthesizing and analyzing existing knowledge (Ioannidis et al., 2015). This growth not only extends the time required to conduct reviews but also raises concerns about the completeness and timeliness of reviews at publication (Shojania et al., 2007). The intersection of AI-ChatBots and SLR represents a novel area of investigation with far-reaching implications for the future of scientific research. Bornmann and Lepori (2024) argued that the rapid evolution of the LLMs quickly renders empirical findings on their use obsolete, emphasizing the need for continuous research in this area. Several studies have explored the application of AI-ChatBots in different aspects of the research process, including article screening (Syriani et al., 2024), topic identification (Tang et al., 2024), search query formulation (Wang et al., 2023; Guimarães et al., 2024), and citation prediction (de Winter, 2024). However, a comprehensive evaluation of AI's role across the full SLR process remains largely unexplored. Some researchers have proposed new SLR variations that combine ChatGPT, one of the most renowned LLM-powered chatbots globally, and human efforts to leverage the strengths of both (Alshami et al., 2023), but mainly the literature concentrates the attention only on ChatGPT, lacking the exploration of other AI-ChatBots available (for more details on the other main AI-ChatBots available, see Section 3.2). As mentioned previously, ChatGPT has been the subject of several studies investigating its potential in academic research. Torres Salinas et al. (2024) explored ChatGPT's transformative role in enhancing bibliometric research methodologies across various stages of academic study, from preparation and consultation to scientific writing. They highlighted ChatGPT's versatility in streamlining data handling, enhancing analytical depth, and facilitating scholarly communication. However, other studies have yielded mixed results. Farhat et al. (2023) found large discrepancies in ChatGPT's performance on bibliometric analysis tasks, cautioning researchers about its trustworthiness in this area.

The potential of AI to address some of the challenges associated with SLRs is significant. AI-ChatBots could potentially speed up the process, addressing the long-recognized limitation of the labor-intensive nature of systematic reviews (Borah et al., 2017). However, the use of AI in research also raises significant concerns. Issues of bias, transparency, and the possibility of error or misinterpretation are paramount. AI may perpetuate biases present in its training data, potentially altering the results of the reviews. One of the most serious problems present in various LLMs, which weighs heavily, especially in SLRs, is the phenomenon of AI hallucinations. AI hallucination is defined as a phenomenon in which an AI system generates a response that is convincing but

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

completely fabricated (Athaluri et al., 2023). The term hallucination has been critically debated lately in the literature, with more *decisive* naming prosed given the harmful nature of this phenomenon (Hicks et al., 2024). For example, it is common for the AI-ChatBots to generate fake *realistic* scientific references. This phenomenon is common to several LLMs and AI-ChatBots, with percentages exceeding 25% of the references generated, as presented by Chelli et al. (2024). The issue of hallucination raises concerns in the academic field about the integrity and accuracy of the use of LLMs in academic writing (Alkaissi and McFarlane, 2023), particularly in rigorous work such as SLRs. The *black box* nature of many AI systems raises issues of transparency and reproducibility, fundamental tenets of scientific research (Lipton, 2018). Moreover, while AI can process and synthesize information on a scale beyond human capabilities, it lacks the depth of understanding and critical thinking that human researchers bring to the review process. The potential misuse of AI in academic writing has also been a point of concern. Haman and Školník (2024) criticized the use of ChatGPT in literature reviews due to the problem of information delivered based on non-existent papers. This highlights the need for careful verification and the maintenance of high ethical standards when integrating AI tools into the research process.

3. Materials and methods

This section reports the method used to do the analysis (including the prompts used in the AI-ChatBots), the various AI-ChatBots used in the analysis next, and the benchmark data used in the comparison between the AI-ChatBots results and human SLR at the end.

3.1. Methods

The work aims to test various AI-ChatBots in the context of the SLR process. SLRs aim to collect all relevant empirical evidence to answer a specific research question (Petticrew and Roberts, 2008). Thus, as discussed in Avenali et al. (2023), SLRs are valuable for collecting meta-knowledge. The method emphasizes transparent and replicable steps to minimize bias and produce more reliable findings. As Avenali et al. (2023) suggested, the main steps of SLRs are:

1. the planning phase, where activities are planned and brainstormed on the topic to be covered;
2. the searching phase, where scientific articles are searched in official and unofficial databases;
3. the screening phase, where articles are selected according to the topic and based on eligibility criteria (typically pre-defined during the planning phase);
4. the data extraction phase, where information is extracted from each article according to the objective of the review; and
5. the reporting phase, where the results of the SLR are presented (usually in the form of a scientific article).

As stated in the Introduction, AI-ChatBots may enhance the SLR process by automating key tasks like screening, eligibility assessment, data extraction, and summarization. They may analyze abstracts, titles, keywords, and full texts to determine relevance, significantly speeding up

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the screening phase. Additionally, they can assist in extracting structured data, reducing manual effort and errors. While they are not yet capable of directly synthesizing findings, they could help organize extracted data and identify patterns. Despite their efficiency and potential bias reduction, human oversight remains essential to ensure accuracy and validity, especially in critical tasks such as article screening.

To study how the different AI-ChatBots can be used in the SLR process, we decided to focus our attention on the case study presented in the paper proposed by Avenali et al. (2023). Specifically, we tested these tools to perform a *simplified* SLR, where the simplification relies on the bibliometric-oriented eligibility criteria for the selection of the articles (30 most cited articles present in *ITOR*) and the non-present screening phase. Previous studies, such as Syriani et al. (2024), have specifically focused on automating the screening phase due to its complexity. This step is not only challenging to automate but also difficult to evaluate objectively, as relevance judgments often require human interpretation. Conducting a detailed study on this task alone would have limited our ability to assess AI-ChatBots across other key SLR phases that are less frequently examined. To ensure methodological rigor and reproducibility, we opted for a benchmark that is already established in the literature (Avenali et al., 2023). This approach allowed us to conduct a transparent comparison of AI-ChatBots without the influence of subjective screening decisions.

For each phase, it is assessed whether the results obtained are acceptable or not. To evaluate the performance of AI in this review, our study employs a two-phase method.

The first phase involves applying *prompt engineering* principles, as outlined by Walter (2024), to develop and refine prompts. These prompts are the inputs or instructions given to an LLM to answer a specific question or guide the model in performing a task. They are typically in the form of text and can range from simple questions or commands to more complex scenarios. Developing and refining prompts is crucial for AI analysis in an SLR because it aims at ensuring accurate and relevant responses while maintaining consistency across queries. Well-structured prompts help guide the AI in interpreting research data effectively, reducing bias, and improving the quality of the summary of findings.

In our study, to create the prompts, we employed the *zero-shot prompting paradigm* for interacting with the AI-ChatBots (Wei et al., 2021). Zero-shot prompting refers to a language model's ability to perform a task without being provided with specific examples for that task. Instead, it relies on its pre-existing knowledge acquired during training on a vast dataset. This contrasts with few-shot prompting (Brown et al., 2020), where the model is given a few examples of the task to guide it and improve its performance, and chain-of-thought prompting (Wei et al., 2022), a technique aimed at enhancing the model's reasoning abilities by encouraging it to generate a series of intermediate steps (a *chain of thought*) leading to the final solution. The choice of using the zero-shot technique relates to its ease of application to the various stages of separate SLR.

The second part consists of a comparative analysis between the outputs generated by AI-ChatBots and a benchmark established through a manually conducted SLR (further details provided in Section 3.3). This benchmark serves as a reference point for assessing AI's performance, aligning with established practices in performance evaluation. Performance evaluation and comparison between different LLMs is a complicated practice. There are different approaches, such as *human-based* evaluation and the use of benchmark databases (Chiang et al., 2024). In our work, we evaluated the results obtained from the AI-ChatBots by comparing them with *human-based* results. For each stage, the evaluation method adopted is different. Below, we detail, for each stage,

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the prompts used and the evaluation method used. It is important to notice that the screening phase is not present because it was not part of the SLR process adopted (more details in Section 3.3).

3.1.1. Planning

In the planning stage, the *Termboard* website (Hoogkamer, 2024) was employed as a tool for brainstorming. It is important to note that we deliberately excluded the use of the AIs for human resource planning within this study, as it falls outside the scope of our current research objectives (though this could be a potential future expansion of this research as done in other fields like event planning) (Keiper, 2023). Brainstorming is a part of the planning phase where the researchers need to generate numerous ideas rapidly on the SLR topic. Thanks to this tool, we generate a prompt to create a conceptual map. Concept mapping is a cognitive tool to produce structural knowledge representation and is particularly effective for brainstorming during the planning phase of an SLR (Jonassen, 1996). This visual technique facilitates the organization of complex information, identification of relationships between concepts, and recognition of knowledge gaps, thereby supporting collaborative thinking among researchers. This approach not only enhances the structural knowledge representation but also promotes critical thinking in the early stages of the review process (Dabbagh, 2001). Regarding the brainstorming topic, the objective was to have AI-ChatBots generate a comprehensive overview of the primary subjects addressed in *ITOR*. The specific prompt employed during this phase was as follows:

> *Do not give any explanations. Use the context of "Main Topics Covered by the Journal International Transactions in Operational Research". Create an ontology on "International Transaction in Operational Research (ITOR)". Create a table of relations with 3 columns in this order: source, target, and relation name. And also create a table with 2 columns: put in the first column the name of the term and in the second column an elaborate definition of the term.*

The prompt was executed on August 5, 2024. Once the answers are obtained, again thanks to *termboard*, they are transformed into conceptual maps. These maps were subsequently evaluated qualitatively based on the terms and relationships found in connection with the presentation of ITOR provided by the official website (ITOR, 2024). Therefore, the evaluation made is based on the ability of the various AI-ChatBots to best express the concepts expressed in the following text:

> *International Transactions in Operational Research is the official journal of the International Federation of Operational Research Societies (IFORS). As such, International Transactions in Operational Research is committed to advance and promote the methodology, understanding, education and practice of Operational Research (OR) throughout the World. International Transactions in Operational Research (ITOR) aims to advance and promote the methodology, understanding, education and practice of Operational Research (OR) by creating bridges between theory and application and academics and practitioners. The journal also has the goal of fostering the interface of Operational Research with related disciplines, such as management science, data science and analytics, computer science, artificial intelligence and machine learning, economics and game theory, simulation and statistics, applied mathematics, mathematical programming and optimization, large scale networks, computer and social networks, among others.*

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The qualitative evaluation method adopted has three possible outcomes:

1. acceptable map and *suitable* AI-ChatBot, if the content presented reflects what the site describes;
2. partially acceptable map and *partially suitable* AI-ChatBot, if the content presented reflects what the site describes, but the map generated makes it difficult to understand the connections between the various topics covered by *ITOR*;
3. unacceptable map and *unsuitable* AI-ChatBot, if the content is incomplete, out of context, or misleading to the journal description.

3.1.2. Searching

For the searching phase, two scenarios were adopted. The first scenario involves the use of AI-ChatBots to directly obtain item information based on our criteria. This scenario is one of the typical cases that a user may face, especially when the researcher is not an experienced user and confuses these AI-ChatBots (which are interfaces for accessing the underlying LLM) with search engines (i.e., interfaces for accessing databases).

In this scenario, the 30 most cited articles according to the various AI-ChatBots were compared with the article presented in the Benchmark data in Section 3.3. Also, again for further control, the digital object identifiers (DOIs) provided by the AI-ChatBots were verified being reference creation (i.e., the invention of an inexistent reference), one of the most common hallucinations in this area (Athaluri et al., 2023). A DOI is a persistent identifier or handle used to uniquely identify various objects like academic articles. The DOI for a document remains fixed over the lifetime of the document, whereas its location and other metadata may change. If the results generated by the AI-ChatBots generate nonexistent DOIs or references that are not present in the benchmark dataset, their results are considered *unacceptable* for this phase, and the AI-ChatBot is unsuitable for this first scenario.

The prompt used in this first scenario was:

> *You are the editor in chief of the journal "International Transactions in Operational Research". You are searching for the 30 most important articles in your journal based on the number of citations. Provide me a table with these articles. Use 4 columns: name of the authors, number of citations, DOI and APA reference*.

The prompt was launched on August 2, 2024.

As a second scenario, we used AI-ChatBots to formulate search queries on academic databases typically used in SLRs (Scopus and Web of Science [WOS]). This type of approach has been studied before for ChatGPT (Wang et al., 2023; Guimarães et al., 2024) but not for other AI-ChatBots. The prompt used in this scenario was:

> *You are the editor-in-chief of the journal "International Transactions in Operational Research." You are searching for the 30 most important articles in your journal based on the number of citations. Provide me with a query to use in Web of Science or Scopus to retrieve these articles*.

The data were retrieved on August 2, 2024. After generating the queries, they were executed in the respective databases to verify their accuracy. The acceptability of the queries was determined based

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on whether they successfully retrieved the intended articles. If both queries were correct, the AI-ChatBot was considered *suitable* for this scenario. If only one query was correct, it was considered *partially suitable*. However, if neither query was correct, the AI-Chatbot was deemed *unsuitable* for this task.

3.1.3. Data extraction

For the data extraction phase, we directly attached the PDF files of the 30 most cited articles from *ITOR*, as identified by Scopus, to the prompt. However, the free versions of the AI-ChatBots analyzed currently allow only one PDF file to be uploaded at a time. Additionally, LeChat does not support PDF uploads, and the free version of Gemini also lacks this capability (unlike its paid version). As a result, both Gemini and LeChat were excluded from this stage. To ensure a fair comparison, we did not use the results from the previous phase when evaluating the AI-ChatBots in this stage. This approach allowed us to assess their performance independently, without any influence from prior outcomes. For each article, we assigned a straightforward task to the AI-ChatBots: extracting the main topic.

This again follows the work carried out by Avenali et al. (2023). The prompt used is as follows:

> *I'll give you a set of PDFs, one by one. Return me an extremely synthetic summary of the main topic in maximum two lines.*

The date of execution of the prompts is August 19, 2024. A result is considered *acceptable* if the topic is correct (i.e., the content extracted is like the one presented in Section 3.1.4). If the AI-ChatBots fails to analyze the file because it exceeds the technical limits of the tools (such as file weight, file type, or text length), the result is considered *partially acceptable* as the AI-ChatBot recognizes its limits and communicates them to the user, avoiding errors. Conversely, a result is considered *unacceptable* if the topic extracted is not correct (i.e., hallucination, imprecision, or other errors). Thus, an AI-ChatBots is considered *suitable* for the task of data extraction if it generates *acceptable* or *partially acceptable* results in less than 50% of the articles considered (in our case, 15), *partially suitable* if it generates *acceptable* or a number of *partially acceptable* results more than 50% of the articles considered, while it is *unsuitable* for this task if *at least one result* is unacceptable.

3.1.4. Reporting

The reporting phase of an SLR is the final stage, where the findings from the review process are summarized, structured, and presented in a coherent manner. An SLR report is a structured document that provides a comprehensive and methodologically rigorous synthesis of existing research on a specific topic. It follows a predefined protocol to systematically collect, analyze, and interpret relevant studies, ensuring transparency, reproducibility, and minimization of bias (Page et al., 2021).

Such reports are widely used in academic, medical, and scientific research to offer an evidence-based overview of a research area. To evaluate this stage, we assessed the ability of various AI-ChatBots to generate reports based on the results of the data extraction phase. Specifically, we examined whether these tools could effectively construct what is traditionally referred to as the *Results* section of an SLR. The data used for this evaluation are presented in Section 3.3 (Table 2), which provides a standardized, high-quality dataset (manually extracted) for all chatbots. This

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ensured consistency and mitigated potential biases or errors that could arise from chatbot-generated data extraction. Although the AI-ChatBots primarily analyzed the information in Table 2, they were not explicitly restricted from leveraging their broader knowledge base to provide contextual enrichment. This approach was designed to reflect real-world research practices, where researchers integrate structured data analysis with existing domain knowledge to develop a comprehensive manuscript. Nevertheless, the primary emphasis remained on the data presented in Table 2 to ensure consistency and reliability in our evaluation. The prompt used in this stage was as follows:

> *You are an academic researcher doing a systematic literature review. You are analyzing the 30 most cited article published in the scientific journal "international transaction operations research". Write the report of this review without subsection or bullet point based on the data presented in the table below. [table omitted to save space, see Table 2]*

The date these prompts were executed by the various AI-ChatBots is August 21, 2024. Assessing the quality of the text generated by the various LLMs available is a very complicated and difficult task. There are various methods for assessing the quality in a quantitative manner of the differences between a generated text and a reference text (e.g. BertScore; Zhang et al., 2019). In our case, given the field of application and the stringency required in SLRs, a manual qualitative assessment was also adopted in this task. Moreover, the aim of our work is, as mentioned above, to understand whether and to what extent these tools can do a task, without creating rankings of the same. To do this, we assessed two aspects: correctness of the information reported (i.e., whether the extracted topic was correctly assigned to the cited article and whether the citations provided were correctly entered) and presence of the correct references of the articles provided. With this type of evaluation, we want to test the reliability of AI-ChatBots in obtaining accurate and complete reports. To do that the evaluation method is as follows: if an AI-ChatBot has made errors in the reported information or has not entered the provided references, the AI-ChatBot is considered *unsuitable* for this task; if it has reported the information correctly and the references provided are correct it is considered a *suitable* tool for this task.

3.2. AI-ChatBots analyzed

The field of AI has seen a significant transformation with the advent of LLMs and their integration into *user-friendly* chatbot interfaces. Many leading companies have released their models for developer use and, subsequently, for general audiences through chatbot interfaces. Each chatbot and its underlying model have their unique characteristics and qualities. For our purposes, we mainly focus on the most important free AI-ChatBots available:

1. ChatGPT (by OpenAI; OpenAI, 2022)
2. Gemini (by Google; Google, 2023)
3. LeChat (by Mistral; Mistral, 2024)
4. Claude (by Anthropic; Anthropic, 2023)
5. CoPilot (by Microsoft; Microsoft, 2023)

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1
Comparison of the main features of the analyzed artificial intelligence (AI)-ChatBots

	Website	Model	Multi modality	Knowledge cutoff	Internet access	PDF reading
ChatGPT	https://chatgpt.com/	GPT-4o	Yes	August 2023	Yes	Yes
Claude	https://claude.ai/new	3.5 Sonnet	Yes	April 2024	No	Yes
LeChat	https://chat.mistral.ai/chat	Large 2	No	October 2023	Yes	No
CoPilot	https://copilot.microsoft.com/?culture=en-us	GPT-4	Yes	October 2023	Yes	Yes
Gemini	https://gemini.google.com/app	1.5 Flash	Yes	November 2023	Yes	No (for free users)

A fundamental concept in comprehending these models is the *knowledge cutoff*, which denotes the date up to which the model has been trained with data and information. Beyond this cutoff date, the model lacks direct access to new information. Each model (and the chatbot that calls it) has a different cutoff, so it is important to know them to know their temporal limitation.

Historically, ChatGPT was the first AI-ChatBots made freely accessible to the public and remains to date the most known and widely used and studied in literature. OpenAI released ChatGPT on November 30, 2022. Up to date, ChatGPT uses the latest model developed by the company, GPT-4o (released on May 13, 2024). The *o* a shortening for *omni*, emphasizing its multimodal capabilities in handling various input and output formats. In our analysis, we employed the free version of ChatGPT with GPT-4o model. Anthropic's Claude was designed with a focus on ethical considerations and security. The most advanced model available through their chat interface is the Claude 3.5 Sonnet model, which was released on June 20, 2024, and was used for the presented analysis. Google's Gemini, released in early 2024, represents the first departure from traditional LLMs architectures with its native multimodal approach. Gemini is available in different versions with the Gemini 1.5 Flash model being the only free option available through Gemini Chat (in August 2024; while the others are accessible through the paid version). The Gemini 1.5 Flash model was employed in the present work. Microsoft's Copilot, integrated into Windows 11 on September 26, 2023, leverages OpenAI's GPT-4 model and uniquely offers direct access to websites. LeChat (by Mistral) takes a different approach by employing multiple specialized models for specific tasks, demonstrating how LLMs architectures can be tailored for specific domains. The model adopted for this article was the latest and most advanced model available: the Large 2 model. These AI-ChatBots differ in their multimodal capabilities, knowledge cutoff dates, Internet access, and PDF files reading abilities as detailed in Table 1. This diverse landscape illustrates the rapid advancements and varied approaches in AI, each offering unique capabilities and reflecting different priorities in their development and application.

3.3. Benchmark data

As already mentioned, in this article, a comparison between the AI-ChatBots is presented. To compare their results, we need reliable *Benchmark* data. To do so, we took over what Avenali et al. (2023) did in their previous SLR. The objective of their SLR was to analyze the main topics of

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the 30 most cited articles published in *ITOR*. Following their example, we adopted their *simple* predefined eligibility criteria based on citations. For the case study, we analyzed the 30 most cited articles according to Scopus of *ITOR* as they did. This choice meant that the screening phase could not be tested, as it was not necessary in the SLR. Furthermore, again for the same reason, the *simplistic* nature of this case study did not need to follow the full protocols normally adopted in SLRs (such as Preferred Reporting Items for Systematic reviews and Meta-Analyses, PRISMA, Page et al., 2021). Consequently, the PRISMA scheme typically adopted in SLRs is not reported as it is unnecessary. A comparison of our results with those reported by Avenali et al. (2023), derived from their Scopus data retrieval, reveals discrepancies in citation counts. Our results indicate an average of 210 citations, compared to 183 citations for the top 30 articles in Avenali et al. (2023). This discrepancy is primarily attributable to the time lag between the respective data retrieval dates. Notably, we observe significant changes among publications with fewer citations in the list. In our analysis, Rojas Viloria et al. (2021), de Freitas and Penna (2020), Narbón-Perpiñá and De Witte (2018b), and Vu et al. (2017) are newly ranked among the top 30 most cited articles. In contrast, Gehring and Bortfeldt (2002), Pérez-Bernabeu et al. (2015), Preux and Talbi (1999), and Scott and Read (1996) have dropped from the top 30. Interestingly, the top 4 positions remain unchanged, with Sörensen (2015) still holding the position of the most cited article in the journal, followed by Laporte et al. (2000), Bana e Costa and Vansnick (1994), and Chaudhry and Khan (2016). Also following their example, the manual extraction of the main topic of each article was performed. Table 2 shows the articles analyzed, the citations reported by Scopus (in date Auguts 10, 2024), the topic extracted, and the Scopus citation reported by Avenali et al. (2023). These data serve as a benchmark for the performance evaluation of the various AI-ChatBots in the searching and data extraction phases and is used as input for the reporting phase.

4. Results

This section reports the main results of the work. The results are presented in subsections, following the order of the SLR process presented in Section 3.1.

4.1. Planning

Figures 1–5 present conceptual maps generated for the planning phase. These maps illustrate a brainstorming process aimed at defining the topics covered by *ITOR*, based on the pre-existing knowledge of various AI-ChatBots. Gemini 1.5 flash (Fig. 1) produced a detailed map effectively representing the key topics addressed by *ITOR*. This map demonstrates *ITOR*'s integration of diverse methodologies and its application across both public and private sectors, while also highlighting its foundations in operational research and management science. In contrast, CoPilot (Fig. 2) generated several sub-concepts without a central node related to *ITOR* and included numerous concepts irrelevant to *ITOR*'s focus (e.g., professional issues encompassing ethics and practices). This renders the map unacceptable and the AI-ChatBot *unsuitable* as the generated content is out of context with the journal description, and the lack of a central, unifying concept could also be considered incomplete in representing the core theme of ITOR. ChatGPT (Fig. 3) generated a map with

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2
Benchmark data. Topic extraction defined by the authors. Scopus citation data from Avenali et al. (2023) is presented in the fourth column. "NA" is used for articles not included in their work. Data retrieved by Scopus in date August 10, 2024

Article	Citations	Main topic (manual review)	Citations from Avenali et al. (2023)
Sörensen (2015)	731	*Novel* metaheuristic methods	575
Laporte et al. (2000)	518	Survey on heuristic on vehicle routing problem (VRP)	492
Bana e Costa and Vansnick (1994)	327	Techniques for cardinal measurement of values	299
Chaudhry and Khan (2016)	322	Survey on flexible job shop scheduling problem	249
Rais and Viana (2011)	274	Survey on operations research in health care	232
Tomasini and Van Wassenhove (2009)	271	Humanitarian logistics	234
Alba et al. (2013)	245	Survey on parallel metaheuristics	207
Sinuany-Stern et al. (2000)	241	Data envelopment analysis and analytic hierarchy process combination proposal	232
Colorni et al. (1996)	218	Heuristics *derived* from nature applied to graph optimization problems	204
Gehring and Bortfeldt (1997)	209	Genetic algorithm (GA) applied to the container loading problem	193
Guajardo and Rönnqvist (2016)	204	Review on cost allocation methods	161
Osman and Christofides (1994)	188	Discussion on capacitated clustering problem	174
Fagerholt (1999)	178	Optimal fleet design in a ship routing problem	165
Malaguti and Toth (2010)	177	Survey on vertex coloring problems	157
Archetti and Speranza (2012)	165	VRPs with split deliveries	148
Van Wassenhove and Pedraza Martinez (2012)	165	Supply chain management (SCM) on humanitarian logistics	145
Rojas Viloria et al. (2021)	159	Literature review on drones in VRPs	NA
Festa and Resende (2009a)	153	Bibliography/survey of greedy randomized adaptive search procedure (GRASP)—Part I	146
Kozan and Preston (1999)	153	GAs to applied to scheduling container problem	145
Yu et al. (2018)	150	Interactive multicriteria decision-making approach proposal	138
Narbón-Perpiñá and De Witte (2018a)	149	Systematic literature review (SLR) on local governments' efficiency	120
Festa and Resende (2009b)	142	Bibliography/survey of GRASP–Part II	135
Van der Vorst et al. (1998)	139	SCM on logistical performance in food supply chains	130
de Freitas and Penna (2020)	137	Variable neighborhood search for flying sidekick traveling salesman problem	NA
Constantin and Florian (1995)	124	Nonlinear bi-level programming approach applied to transit networks	108
Yeh (2002)	121	Selection of compensatory multi-attribute decision-making methods	109
Narbón-Perpiñá and De Witte (2018b)	116	SLR on local governments' efficiency (Part II)	NA

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2
(Continued)

Article	Citations	Main topic (manual review)	Citations from Avenali et al. (2023)
Mingers (2000)	110	Discussion on soft methods and their combinations.	96
Vu et al. (2017)	109	Surrogate-based methods for block-box optimization	NA
Liberti et al. (2008)	109	Branch-and-prune algorithm for the molecular distance geometry methods	104

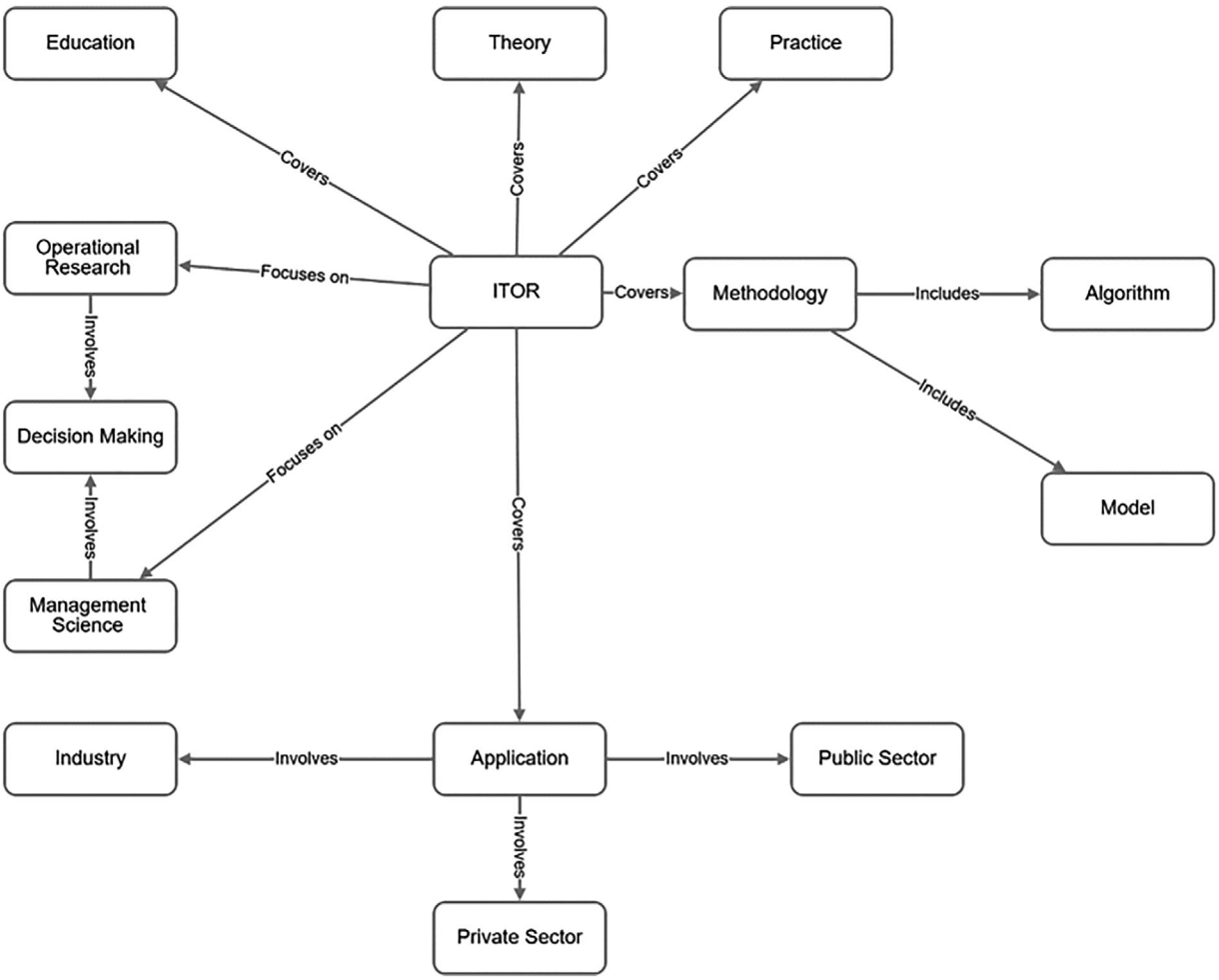

Fig. 1. Conceptual map of *International Transactions in Operational Research* (ITOR) derived from the ChatBots' pre-existing knowledge, generated by Gemini (Flash 1.5). The first map was generated on June 19, 2024, and updated on August, 5, 2024.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

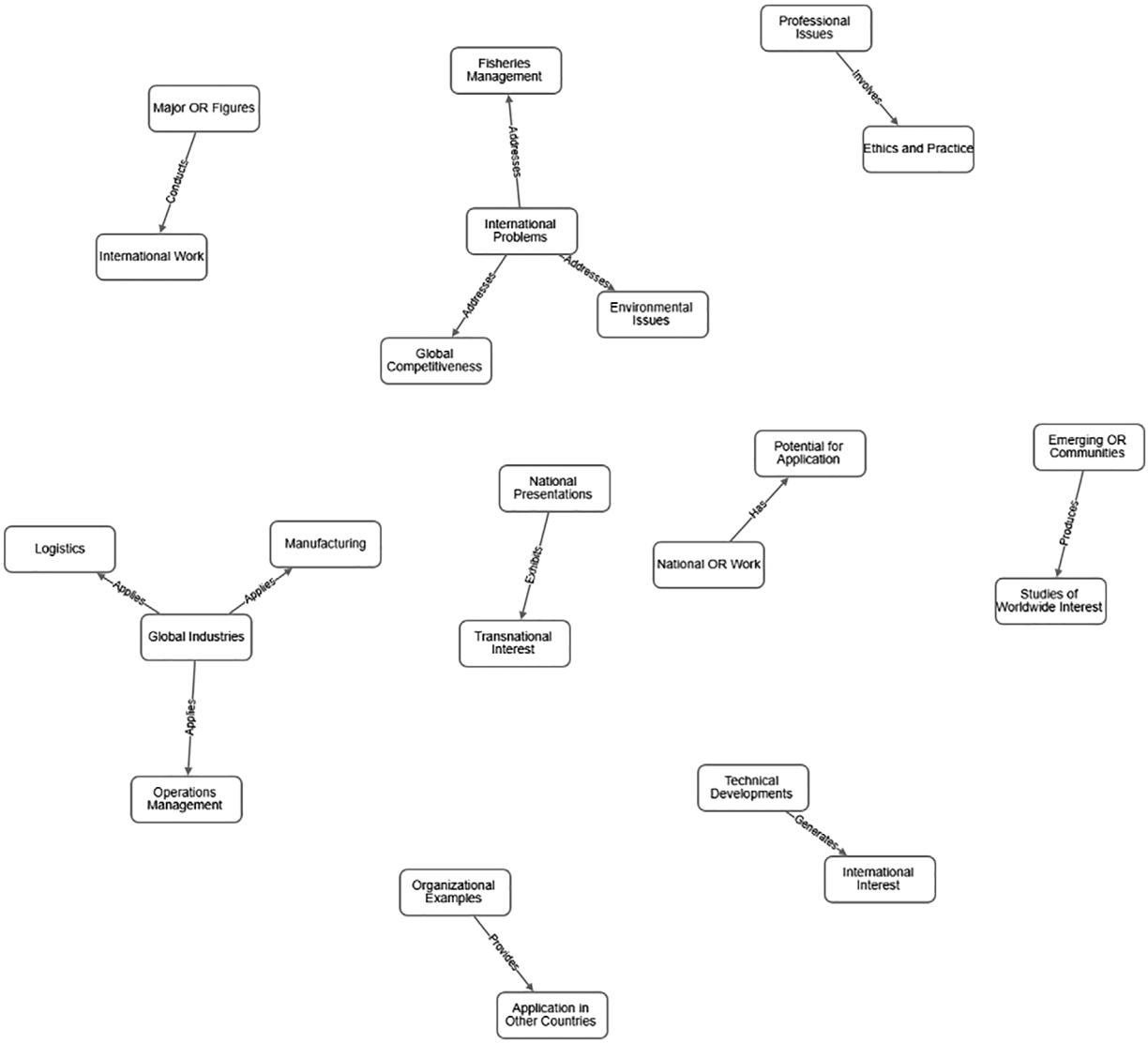

Fig. 2. Conceptual map of *ITOR* derived from the ChatBots' pre-existing knowledge, generated by CoPilot (GPT-4). The first map was generated on June 19, 2024, and updated on August, 5, 2024.

a unique *two-concept* structure. Although it does not present *ITOR* as the root or central concept, it details the journal's main topics. While the presented concepts are accurate, this map's unusual structure is only partially suitable for brainstorming, as it obscures the connections between the various topics covered by *ITOR*. Claude (Fig. 4) provided the most comprehensive overview of the journal, compared to the other AI-ChatBots, presenting not only core concepts (e.g., optimization and simulation) but also related concepts for a deeper understanding of the topics covered. Finally, LeChat (Fig. 5) generated a concise but accurate representation of *ITOR*. Comparing the results, the maps generated by Gemini 1.5 flash and Claude were deemed the most effective, followed by

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

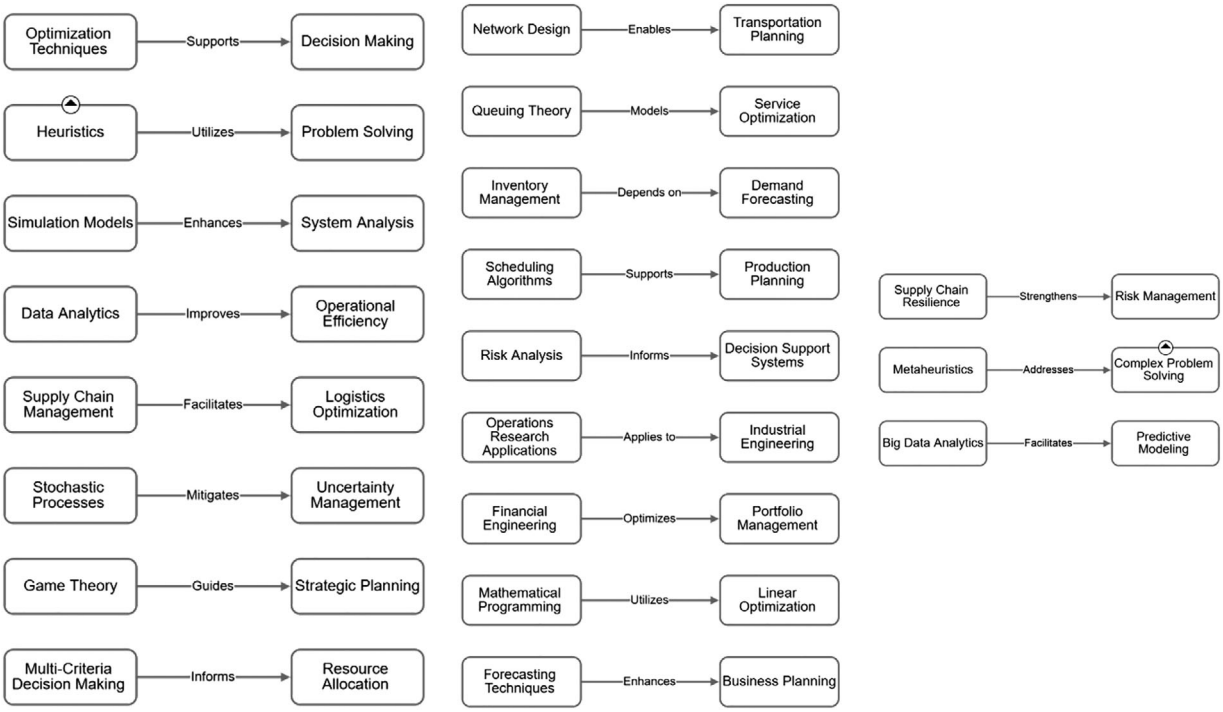

Fig. 3. Conceptual map of *ITOR* derived from the ChatBots' pre-existing knowledge, generated by ChatGPT (GPT-4o). The first map was generated on June 20, 2024, and updated on August, 5, 2024.

Table 3
AI-ChatBots-generated concept maps evaluations.

ChatBot (model)	Planning (brainstorming)
Gemini (1.5 Flash)	Suitable
Copilot (GPT-4)	Unsuitable
ChatGPT (GPT-4o)	Partially suitable
Claude (3.5 Sonnet)	Suitable
LeChat (Large 2)	Suitable

an *acceptable* map from LeChat and a *partially acceptable* map from ChatGPT. The map generated by CoPilot was considered *unacceptable*. Therefore, Gemini, Claude, and LeChat are *suitable* for brainstorming; ChatGPT is *partially suitable*; and CoPilot is *unsuitable*. A summary of the AI-ChatBot evaluation for the brainstorming phase is presented in Table 3.

4.2. Searching

The first scenario of the search phase assessed the AI-ChatBots' ability to handle requests they cannot fulfill. Here, the chatbots were divided into two groups. The first group, consisting of

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

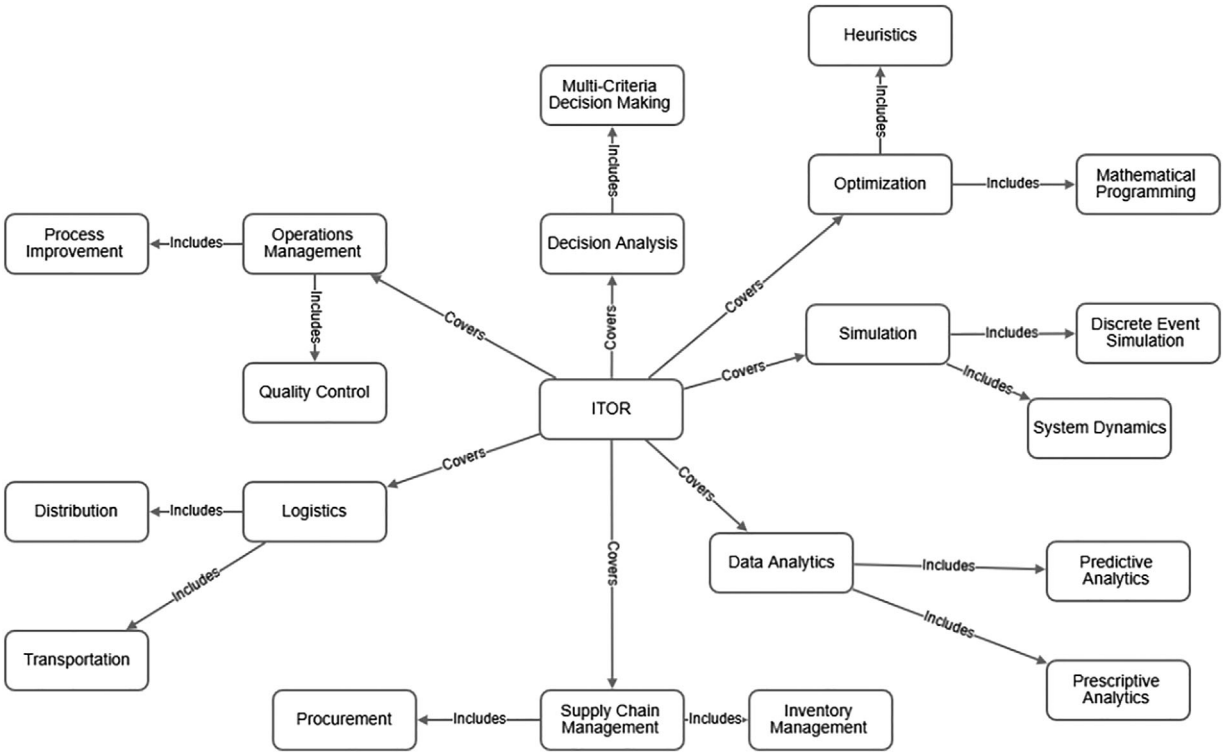

Fig. 4. Conceptual map of *ITOR* derived from the ChatBots' pre-existing knowledge, generated by Claude (3.5 Sonnet). The first map was generated on June 20, 2024, and updated on August 5, 2024

Gemini, Claude, and LeChat, acknowledged their limitations and informed the user that they could not retrieve the requested information (with a message such as *I am unable to access Web of Science or Scopus to retrieve the requested articles* or clarifying that they *do not have direct access to proprietary databases like Web of Science or Scopus, which are subscription-based services*). They also indicated that any tables they could provide would be for illustrative purposes only and explicitly stated that the data were not accurate. This behavior demonstrates that these AI-ChatBots have safeguards in place to identify requests beyond their capabilities and to communicate these limitations to the user. In contrast, ChatGPT and CoPilot provided the requested information, even though with estimations and acknowledgments that the data might not be precise or up to date. It is important to consider the likely source of this information. Although ChatGPT and CoPilot provided this information, their caution about accuracy suggests that they did not directly access proprietary databases such as Scopus or WOS. It is unclear whether they drew on their extensive training dataset, containing metadata and references from open sources, or whether they used Web search features to retrieve information from public Web pages. Without specific details provided by the ChatBot on the query process, it is difficult to determine with certainty the source for these data. The query results are presented in Appendix A (Table A1).

In terms of correctness, only CoPilot found one *ITOR* article with a correct title and DOI, but it mistakenly identified the author. All the other ChatBots have incorrect DOI or

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

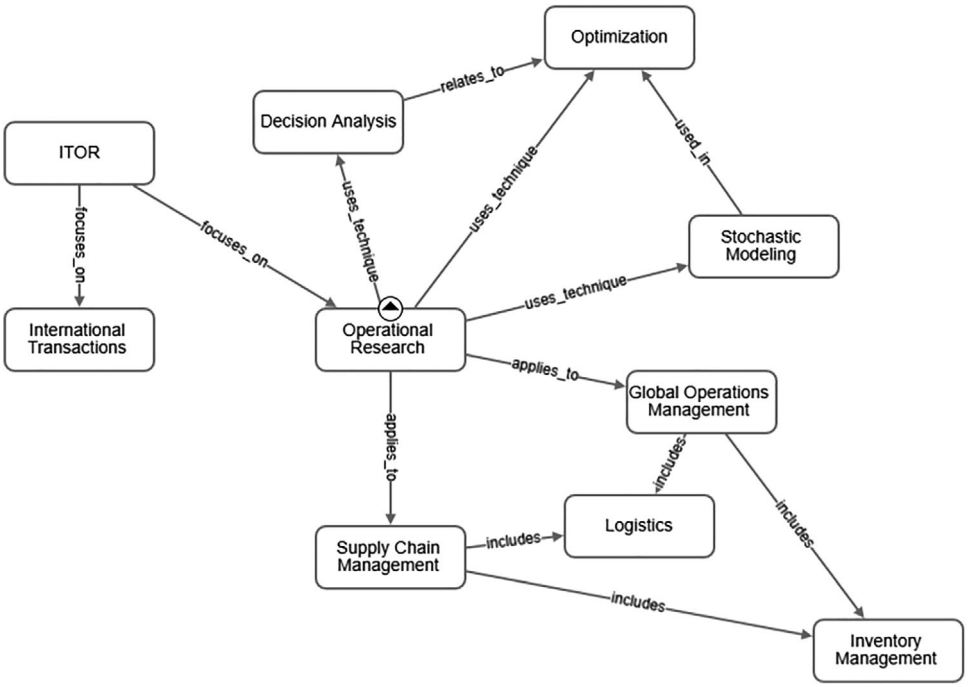

Fig. 5. Conceptual map of *ITOR* derived from the ChatBots' pre-existing knowledge, generated by LeChat (Large 2). The first map was generated on June 20, 2024, and updated on August 5, 2024.

mistakenly indicated the article. However, an interesting and concerning result emerged: CoPilot generated realistic DOIs with the correct *ITOR* prefix (i.e., having the first part of the https://doi.org/10.1111/itor). Nevertheless, these DOIs directed to existing *ITOR* articles that did not correspond to the expected publications. ChatGPT, on the other hand, generated non-existent DOIs, and none of the provided articles belonged to *ITOR*. Many of the articles found were from other journals (e.g., the generated reference "Kusiak, A., & Heragu, S. S. (1987). The facility layout problem. International Transactions in Operational Research, 12(4), 381–396. https://doi.org/10.1111/j.1475-3995.1987.tb00409.x" is incorrect because the article exists but is from the *European Journal of Operational Research* and the DOI is not real).

These results suggest that the safeguarded approach of the first group of AI-ChatBots avoids the core issue, while the request-responsive chatbots (CoPilot and ChatGPT) fail this task. The primary reason for their failure is that these chatbots lack access to official databases and, despite potential Internet access, are not designed for comprehensive online searches. Consequently, these two chatbots are *unsuitable* for this scenario.

The second scenario results (the query generated by the AI-ChatBots) are presented in Table 4. ChatGPT generated correct queries initially, but the included *SORT BY* conditions are not supported by either database. CoPilot instead added a condition on the year of publication not specified in the prompt. Interestingly, CoPilot also found the SCOPUS ID (unique identifier for *ITOR* in Scopus) without explicit instructions in the prompt. Despite these additions, both queries were fundamentally correct. LeChat generated accurate queries for WOS but not for Scopus.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4
Queries generated by the different AI-ChatBots for Scopus and Web of Science (WOS) search. Date of retrieval August 2, 2024

ChatBot (model)	WOS	Scopus
ChatGPT (GPT-4o)	SO = ("International Transactions in Operational Research") SORT BY: Times Cited—Highest to Lowest	SOURCE-TITLE("International Transactions in Operational Research") SORT BY: Cited by (highest)
Copilot (GPT-4)	SO = ("International Transactions in Operational Research") AND PY = (1994–2024)	SOURCE-ID(9700153238) AND PUBYEAR > 1993
LeChat (Large 2)	SO = (International Transactions in Operational Research)	SOURCE-ID("International Transactions in Operational Research")
Claude (3.5 Sonnet)	SO = "International Transactions in Operational Research"	SRCTITLE("International Transactions in Operational Research")
Gemini (1.5 flash)	TS = (International Transactions in Operational Research) AND DOCUMENT TYPE(Article)	TITLE("International Transactions in Operational Research") AND DOCTYPE(ar)

Claude successfully generated queries for both databases. Conversely, Gemini 1.5 flash failed in both instances. Based on these findings, Claude emerged as the most capable chatbot, generating correct queries for both Scopus and WOS. CoPilot and ChatGPT generated *partially acceptable* queries due to their aforementioned shortcomings. LeChat's success with WOS queries but failure with Scopus queries relegates it to *partially suitable* as well. Finally, Gemini's (1.5 flash) consistent failure in query generation renders it *unsuitable* for this task.

4.3. Data extraction

In the data extraction phase, CoPilot and ChatGPT proved *suitable* for the task, while Claude was only *partially suitable*. Detailed results of the data extraction phase, due to their length, are presented in Appendix B (Table B1). While all three chatbots extracted topics, their performance varied in completeness. ChatGPT successfully extracted *correct* topics for all 30 articles, whereas CoPilot extracted topics for 29 out of 30 articles. Claude, however, failed to extract topics for 20 articles. This limitation in Claude, at least in its free version, stems from a maximum word limit for parsing, which many of the analyzed articles exceeded. One potential workaround to this limitation would be to divide the articles into sections for analysis. However, this would require downloading, reading, and manually segmenting the PDF files, negating the productivity benefits of using AI-ChatBots and making the process slower than manual extraction. It is important to note that these tests were conducted using the free versions of the AI-ChatBots, and this limitation in Claude may be absent in its paid version.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 5
Final evaluation of the AI-ChatBots' performance in the systematic literature review (SLR) process.

ChatBot (model)	Planning (brainstorming)	Search—Scenario 1	Search—Scenario 2	Data extraction	Reporting
ChatGPT (GPT-4o)	Partially suitable	Unsuitable	Partially suitable	Suitable	Unsuitable
CoPilot (GPT-4)	Unsuitable	Unsuitable	Partially suitable	Suitable	Suitable
LeChat (Large 2)	Suitable	Not available	Partially suitable	Not available	Unsuitable
Claude (3.5 Sonnet)	Suitable	Not available	Suitable	Partially suitable	Unsuitable
Gemini (1.5 flash)	Suitable	Not available	Unsuitable	Not available	Unsuitable

4.4. Reporting

In the reporting phase, CoPilot was the only tool capable of correctly citing all given articles and providing full text. ChatGPT correctly cited 25 of 30 articles, Claude cited 10 of 30 (with one citation error), and LeChat cited 12 of 30 articles. Gemini 1.5 flash completely failed this task, providing uncited and overly general text unsuitable for an SLR. Further details on the generated reports can be found in the *Reports Generated by the AI-ChatBots* Section in Appendix C, which contains the full reports due to their length. Table C1 in the same Appendix provides a detailed list of articles not properly cited in each generated report.

5. Discussion: benefits and pitfalls of AI-ChatBots in the SLR process

From our analysis, no AI-ChatBot proved *suitable* for all phases of the process following the suitability rules presented in Section 3.1 (Table 5). In the brainstorming stage, most AI-ChatBots (except CoPilot and ChatGPT) were effective in the exploratory phase for generating ideas, aligning with Marzuki et al. (2023), who highlighted their utility in student education.

In the search phase (Scenario 1), our findings corroborate known issues such as the reference creation problem (Athaluri et al., 2023) and, more critically, the DOI generation problem (Section 4.2). The latter is particularly concerning, as DOIs are intended to provide persistent and unique identification of scholarly resources. Within the framework of a rigorous methodology like an SLR, such inaccuracies are unacceptable. While this study did not explicitly examine the broader concept of *ethical manipulation* as defined by Klenk (2024), our analysis of AI-generated DOIs revealed significant inaccuracies and inconsistencies (Section 4.2). These findings align with concerns in the literature regarding potential biases embedded in the training data of AI-ChatBots (Glikson and Woolley, 2020; Akgun and Greenhow, 2022). Although further research is needed to determine the root causes of these inaccuracies, our results highlight the critical importance of rigorous validation of AI-generated content in SLR processes. In line with the recommendations of Alkaissi and McFarlane (2023), implementing fact-checking systems to mitigate misinformation is essential. Another potential precaution, already adopted by models such as Claude, Gemini, and LeChat, is the practice of abstaining from generating responses when the model recognizes its inability to provide accurate information. These AI-ChatBots (i.e., Claude, Gemini, and LeChat)

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

acknowledge their limitations and take a precautionary approach by explicitly stating their inability to fulfill a request, thereby avoiding the generation of false outputs. Our findings strongly suggest that researchers should exercise extreme caution when using AI-ChatBots in this specific context and avoid relying on them for tasks where accuracy and reliability are central. Instead, in the Scenario 2 of the searching phase, we find the (partial) possibility of using Claude, ChatGPT, CoPilot, and LeChat to create search queries with results like those shown in past literature (Wang et al., 2023; Guimarães et al., 2024). In the data extraction phase, ChatGPT and CoPilot are shown to be able to perform this task. Considering that normally this task, along with the screening phase, is the most time-consuming part of an SLR, the ability to employ these tools to greatly accelerate this phase brings an important benefit for researchers involved in SLR. At the same time, technical limitations of the limited free versions of Claude and Gemini 1.5 flash arise (the latter not allowing analysis of PDF files in the free version). This kind of issue is a pitfall for resource-scarce academics and researchers from developing countries (Alordiah, 2023) who want to use these tools: To use the paid versions that overcome these limitations, they would have to have a budget available for each paid version of the selected AI-ChatBots (with the exception of LeChat, which currently lacks a paid version). In the reporting phase, focusing on the accuracy of what is reported and not on its form and structure (which depends on the specific cases), CoPilot has proven to be the only accurate and potentially usable tool for this task. The reports obtained at this stage, however, were all found to be credible in their composition (although not always all correct) highlighting a difficulty in recognizing human-generated texts as evidenced previously by Dergaa et al. (2023). The literature in the past has not dwelt much on the reporting phase because of the aforementioned limitations (Bolaños et al., 2024), focusing more on the ability of academic-level abstract creation from the full-text article(s) (Babl and Babl, 2023; Gao et al., 2023). Our results are in line with what has been suggested by Feuerriegel et al. (2023), who pointed out that the possibility of producing seemingly new content from existing data is a revolution in the way we communicate. In addition, the ability to generate an entire draft of a report with a few seconds brings significant benefit to the SLR process (although care should always be taken and time spent to evaluate the accuracy and completeness of what is reported by the AI-ChatBots). An alternative possibility to the use of AI-ChatBots in the reporting phase is their use as writing support tools to expand vocabulary and improve the coherence and logical flow of written texts (Marzuki et al., 2023). The results suggest that AI-ChatBots may contribute to increased efficiency SLR by accelerating some SLR core tasks. Specifically, these tools facilitate the brainstorming of ides (with the noted exceptions of CoPilot and ChatGPT), the generation of search queries for academic databases (considering Claude, ChatGPT, CoPilot, and LeChat), the acceleration of data extraction procedures (especially using ChatGPT and CoPilot), and the potential automation of report drafting, aligning with recent work by Marquis et al. (2024). At the same time, several pitfalls must be considered and carefully evaluated. The main pitfall concerns the *reproducibility* of the review (which is the basis of SLR). The AI-ChatBots demonstrated varying degrees of ability to respond to the provided prompts but may struggle to provide adequate responses when the prompt is unclear or ambiguous (Lee et al., 2023). To further emphasize the importance and impact of prompts and their *understanding*, additional analyses of chatbots' prompt sensitivity are presented in Section 6. Always in the context of *full reproducibility* of the reviews, the process used in SLRs must be transparent and explainable, a problem that is present in the field of AI, however, and which also limits its widespread use in other areas (Dwivedi et al., 2023). Another major pitfall that needs to be considered relates directly

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to the very logic for using these tools in the SLR process: *productivity* improvement. Indeed, from the results obtained, careful manual control (and various preparatory manual steps to their use) is necessary. The impossibility of being able to blindly rely on these tools, without having to subsequently spend time on careful manual checks, goes to impact the time spent by researchers in an SLR. Each researcher should therefore weigh first whether the time spent in using these tools and their subsequent control is worthwhile and less than a manual process. In addition to all the pitfalls indicated, there are also several ethical problems already highlighted in the literature: fairness, security, harmful content (Hagendorff, 2024), plagiarism (Meyer et al., 2023), and copyright law infringement (Delacroix, 2023). The ethical discussion on the use of AI is varied and extensive in the literature, but we limit ourselves to pointing out only these limitations since it is not the focus of this article. Critically, our findings underline the importance of a *human-in-the-loop* approach to maximize the potential benefits and reduce the pitfalls in using AI-ChatBots in SLR (as suggested also by Syriani et al., 2024). By integrating human expertise with AI-ChatBot functionalities, this framework enables iterative refinement and validation, ultimately leading to more robust and reliable review outcomes. All the benefits and pitfalls discussed above, based on the analysis performed according to the rules defined in Section 3.1, are presented in Table 6.

6. Threats to the validity of the findings

As highlighted in the previous section, integrating AI-ChatBots into SLRs offers numerous advantages but also presents significant challenges. One of the key concerns is the *quick obsolescence* of AI models—referring to their rapid evolution through frequent updates, new functionality, and patches. This continuous development makes it difficult to maintain consistent control over the specific model version used, posing a challenge to the reproducibility of findings. To illustrate this issue, we conducted an analysis on January 16, 2025, comparing the performance of two versions of the Gemini model: Gemini 1.5 flash and the newer Gemini 2.0 flash-experimental (see Appendix D). The comparative evaluation of these models across all SLR phases using the same rules presented in Section 3.1 is summarized in Table 7. The results reveal significant variations in Gemini's suitability, with the newer version demonstrating enhanced capabilities across all tested SLR phases. Notably, Gemini 2.0 flash-experimental proved suitable not only for brainstorming but also for searching (Scenario 2) and reporting, underscoring the impact of model updates on both query interpretation and response generation.

This same limitation applies to functionality. As previously noted, on August 2, 2024, LeChat did not support PDF uploads and was therefore excluded from evaluation in this phase. However, by January 16, 2025, this functionality had been introduced, allowing LeChat to be tested. It achieved a perfect score across all articles, demonstrating its effectiveness for this task (Appendix D, Table D1) (Table D2) (Fig. D1).

To ensure comparability with previously reported results, the outcomes from August 2024 have been maintained as the primary benchmark, while the findings from the updated models are presented in a dedicated section.

Furthermore, the sensitivity of AI-ChatBots to specific words and the overall clarity of prompts presents a significant challenge. The tested ChatBots demonstrated varying abilities to interpret and answer to the given prompts. However, their effectiveness can be compromised if they fail to

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 6
Benefits and pitfalls of AI-ChatBots in the SLR process

Phase of SLR	Benefits	Pitfalls
Brainstorming	Almost all AI-ChatBots (excluding CoPilot and ChatGPT) are suitable for generating and do brainstorming on research ideas	None explicitly mentioned for this phase besides exclusion of specific chatbots
Searching (Scenario 1)	None	Reference and digital object identifier generation problems Concerns about biases in training data Requires rigorous validation of AI-generated content
Searching (Scenario 2)	Claude, ChatGPT, CoPilot, and LeChat can create search queries for Scopus and WOS	Possible hallucination on query field search for some ChatBots
Data extraction	ChatGPT and CoPilot can perform data extraction, potentially accelerating this time-consuming phase	Technical limitations of free versions of Claude and Gemini
Reporting	CoPilot is potentially usable for generating reports, offering the possibility of producing seemingly new content from existing data and generating entire report drafts quickly Can be used as a writing support tool for vocabulary expansion and improved coherence	Reports may not always be entirely correct, with difficulty in recognizing human-generated text Not all the ChatBots cited all the references and information provided (information completeness problem) Requires careful evaluation of accuracy Limited prior literature on AI-ChatBots use in the reporting phase
General	Possible Improvement of efficiency in SLR, especially if used in *human-in-the-loop* approach	Hallucination problems, technical limitations, difficulty responding to unclear prompts Reproducibility and transparency of the SLR process problem Requires careful manual control and checks, potentially impacting the time saved. Ethical concerns

fully understand the query (Lee et al., 2023). Even minor wording changes can lead to substantial differences in output, highlighting the critical role of *prompt engineering* in ensuring reliable and consistent results. This sensitivity is particularly evident in the comparison of five distinct prompts used during the search phase (Scenario 2) across various AI-ChatBots (see Appendix E) (Table E1) (Table E2). Additionally, repeating the same prompt at a later date (on January 16, 2025) led to significant variations in the results.

When the same prompts were tested again on January 16, 2025, the results varied significantly; unlike the outcome on August 2, 2024, CoPilot failed to generate both queries correctly across different prompts. These inconsistencies highlight the necessity of precise prompt design and iterative refinement to minimize bias and improve response quality. However, minor variations do not always lead to drastic changes in results. For instance, in the data extraction phase, we considered the prompt including the term *synthetic* and used it to ask for a summary. This word *synthetic* could potentially bias the ChatBot toward generating artificial responses. To assess this impact, we tested

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 7
Evaluation of the Gemini models (1.5 flash and 2.0 flash-experimental) performance in the SLR process.

ChatBot (model)	Planning (brainstorming)	Search—Scenario 1	Search—Scenario 2	Data extraction	Reporting
Gemini (1.5 flash)	Suitable	Not available	Unsuitable	Not available	Unsuitable
Gemini (2.0 flash-experimental)	Suitable	Not available	Suitable	Not available	Suitable

the same prompt on the LeChat ChatBot, replacing *synthetic* with s*hort*. The results remained consistent, indicating that in some cases, slight wording adjustments may not significantly affect the ChatBot's output.

Aware of these challenges, AI developers are actively working to overcome these limitations by introducing new functionalities, tools, and platforms. One such innovation is NotebookLM (https://notebooklm.google.com/), an experimental AI-powered notebook. Unlike traditional AI-ChatBots, NotebookLM allows users to upload their own sources and interact with the documents directly. While current limitations exist regarding the number of sources that can be uploaded, this user-centric approach offers greater control over the AI's knowledge base. This enhanced control has the potential to improve the relevance and reliability of AI-generated information in SLR tasks such as data extraction and reporting. Additionally, NotebookLM includes a podcast functionality that could be an interesting avenue for future research alongside the tool itself.

7. Conclusion

In this paper, we analyzed the potentialities and pitfalls of different AI-ChatBots in guiding and supporting researchers in the SLR process by providing new empirical evidence. To do that, a systematic evaluation of the performance of AI-ChatBots in the main stages of the SLR process and a comparison with a manual SLR was presented. Despite the reproducibility challenges discussed in Section 6, this study provides valuable insights and a foundational framework for evaluating the suitability of AI-ChatBots in the SLR process, highlighting both their benefits and limitations. While the specific findings may become outdated due to the rapid evolution of LLMs, the overarching concept of integrating AI into various SLR tasks, as well as the proposed methodology for assessing chatbot performance across these phases, remains relevant and widely applicable.

Our results indicate that no single *best* AI-ChatBot exists among those analyzed; rather, as detailed in Table 5, each chatbot exhibits distinct capabilities and is therefore better suited to specific tasks. This study identified several potential applications of AI-ChatBots within the SLR process. For initial brainstorming, LeChat, Claude, and both the 1.5 and 2.0 flash-experimental versions of Gemini demonstrated promising results. Claude and the Gemini 2.0 flash-experimental proved suitable for generating search queries for databases such as Scopus and WOS. CoPilot and ChatGPT emerged as appropriate tools for data extraction, while CoPilot and Gemini 2.0 flash-experimental were also well-suited for generating the final report. However, it is crucial

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to acknowledge the identified pitfalls, including instances of hallucination and inaccuracies in DOI/reference creation (as illustrated in Scenario 1 of the search phase in Section 4.2), concerns regarding the reproducibility and transparency of the research process, the need for substantial manual oversight of AI-generated outputs, and various ethical considerations. These numerous challenges emphasize the necessity for careful and judicious use of these tools in SLR.

Considering the prompt used in this study, the research used *zero-shot* prompts techniques. While easy to adapt for the different SLR tasks, this *zero-shot* approach may influence the results. Future research should explore more advanced prompting techniques, such as *few-shot* and *chain-of-thought* prompting, which have the potential to improve the robustness and reliability of chatbots answers. The variations in results observed in Section 6, along with the influence of different prompt techniques and wording, suggest a promising new research direction on developing *appropriate prompts* tailored to specific SLR tasks. These prompts would need continuous refinement to adapt to updates in LLMs. However, this investigation falls beyond the scope of the present study.

Furthermore, future research should evaluate new models and functionalities continuously being implemented by AI-ChatBots companies. This includes the rigorous testing of *deep search* capabilities (such as the OpenAI version https://openai.com/index/introducing-deep-research/ or the Google version https://gemini.google/overview/deep-research/), which can be defined as functionalities that move beyond simple keyword matching to enable a deeper semantic understanding and contextual analysis of information within documents, potentially uncovering more relevant insights (very useful in the SLRs context). Additionally, investigations should explore the application of *thinking* models (such as Gemini 2.0), referring to AI models designed to enable more complex reasoning, planning, and step-by-step problem-solving processes, potentially mimicking human cognitive pathways in synthesizing information. Evaluating the performance of these advanced models and features within the various stages of an SLR will be essential to fully understand their potential impact and limitations. The inconsistent results from identical prompts at different times (as discussed in Sections 5 and 6) pose a threat to the validity of these findings and research that use these tools in SLR in general. This variability undermines the full reproducibility principle of AI-driven SLRs and jeopardizes the fundamental *systematic* nature of the SLR process. This limitation is particularly critical in rigorous research fields such as medicine, where full reproducibility and adherence to rigorous protocols like PRISMA in SLRs are mandatory.

Beyond the aforementioned challenges, several other research limitations need to be emphasized.

The benchmark review employed was *simplistic* in its selection criteria, preventing the evaluation of the article screening stage. This simplification was necessary for our investigation to have an *objective* comparison set to compare the performance of the different AI-ChatBots. However, this simplification highlights the need for further research employing a more comprehensive and realistic SLR. While the current study provides an initial exploration, future work should expand the scope to include a more complete SLR process, encompassing all typical stages. This would allow for a complete evaluation of AI-ChatBot performance. This more complete analysis should also address the critical challenge of identifying and excluding publications from predatory publishers (see, e.g., Al-Moghrabi et al., 2024). This is a significant concern in academic research, and integrating this aspect into future studies would provide valuable insights into the potential of AI-ChatBots to enhance the rigor and quality of literature reviews.

Second, the limited scope of the case study and the benchmark data employed restrict the generalizability of the findings, suggesting future research should encompass diverse academic fields.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Developing official SLR databases would enable more robust benchmarks and systemic evaluations, facilitating the creation of a performance observatory like the Chatbots Arena proposed by Chiang et al. (2024). This would also allow for broader analyses encompassing other AI-based tools such as Perplexity (Perplexity AI Inc., 2022), AI-notebooks (like Notebook-LM or ChatGPT canvas), and the premium chatbot versions (like Gemini Pro) or other LLMs (such as DeepSeek, https://www.deepseek.com/). Another limitation of the research is the focus only and exclusively on free AI-ChatBots, excluding versions and premium features. Additionally, the evaluation of these chatbots is confined to the SLR context, look after their potential use in other SLRs tasks such as text translation or grammar assistance. Future work should explore these tools, which could significantly benefit researchers, particularly non-native English speakers.

Finally, this study did not address the ethical implications of AI use in research, a topic of growing concern among academic publishers and observers. While AI itself is not inherently problematic, its use and potential misuse by researchers raise important ethical and academic integrity considerations.

In conclusion, therefore, AI-ChatBots may play a key role in SLRs in the future, but at present, not all AI-ChatBots can perform this role. Future advancements in AI-driven SLRs offer the potential for increased time-efficiency; however, this potential gain in speed carries the risk of diminished result quality due to the presented pitfalls. Furthermore, the use of fully AI-generated SLRs as training data for new LLMs presents a significant concern. Such a practice could propagate these inaccuracies, leading to entirely *hallucinated* reviews and potentially jeopardizing the integrity of academic research. Considering the rapid evolution that various companies have had in recent months in creating increasingly powerful and accurate models, this future may not be that far off. The diffusion of these tools in research, however, is precisely this rapid evolution: updated builds of the various LLMs render research obsolete in weeks (if not days) and require constant (and laborious) continuous update on the various builds by researchers. Concurrently, various scientific publishers are pushing for a future of full AI-SLR. This push is manifested through commercial agreements wherein publishers sell scientific articles to companies for model training (like recently done by Wiley and Taylor & Francis) and through the implementation of AI-ChatBots trained on official academic databases, (such as Elsevier's *Scopus AI* or the AI project of Clarivate *Primo Research Assistant*).

As such, future research should focus not only on how to use these AI-ChatBots but also on how to regulate their use and how to make them safe, accessible and how it is possible to overcome the presented pitfalls.

Acknowledgments

The author Cinzia Daraio is a member of the board of *International Transactions in Operational Research*. We are deeply grateful to three anonymous reviewers for their deep and competent comments and suggestions that allowed us to greatly improve the quality of our paper. During the preparation of this work, the authors used ChatGPT, Gemini, Claude, LeChat, and CoPilot for the analysis. The use of these tools in the paper is fully detailed and shown. The author takes full responsibility for the content of the publication.

Open access publishing facilitated by Universita degli Studi di Roma La Sapienza, as part of the Wiley - CRUI-CARE agreement.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References

Adamopoulou, E., Moussiades, L., 2020. Chatbots: history, technology, and applications. *Machine Learning with Applications* 2, 100006.

Akgun, S., Greenhow, C. 2022. Artificial intelligence in education: Addressing ethical challenges in K-12 settings. *AI and Ethics*, 2(3), 431–440.

Akpan, I.J., Kobara, Y.M., Owolabi, J., Akpan, A.A., Offodile, O.F., 2025. Conversational and generative artificial intelligence and human–chatbot interaction in education and research. *International Transactions in Operational Research* 32, 3, 1251–1281.

Alba, E., Luque, G., Nesmachnow, S., 2013. Parallel metaheuristics: recent advances and new trends. *International Transactions in Operational Research* 20, 1–48.

Alkaissi, H., McFarlane, S.I., 2023. Artificial hallucinations in ChatGPT: implications in scientific writing. *Cureus* 15, 2, e35179.

Al-Moghrabi, D., Arqub, S.A., Maroulakos, M.P., Pandis, N., Fleming, P.S., 2024. Can ChatGPT identify predatory biomedical and dental journals? A cross-sectional content analysis. *Journal of Dentistry* 142, 104840.

Alordiah, C.O., 2023. Proliferation of artificial intelligence tools: adaptation strategies in the higher education sector. *Propellers Journal of Education* 2, 53–65.

Alshami, A., Elsayed, M., Ali, E., Eltoukhy, A.E.E., Zayed, T., 2023. Harnessing the power of ChatGPT for automating systematic review process: methodology, case study, limitations, and future directions. *Systems* 11, 351.

Anthropic, 2023. Claude. Available at https://claude.ai/new (accessed 28 August 2024).

Archetti, C., Speranza, M.G., 2012. Vehicle routing problems with split deliveries. *International Transactions in Operational Research* 19, 3–22.

Athaluri, S.A., Manthena, S.V., Kesapragada, V.S.R.K.M., Yarlagadda, V., Dave, T., Duddumpudi, R.T.S., 2023. Exploring the boundaries of reality: investigating the phenomenon of artificial intelligence hallucination in scientific writing through ChatGPT references. *Cureus* 15, 4, e37432.

Avenali, A., Daraio, C., Di Leo, S., Matteucci, G., Nepomuceno, T., 2023. Systematic reviews as a metaknowledge tool: caveats and a review of available options. *International Transactions in Operational Research* 30, 2761–2806.

Babl, F.E., Babl, M.P., 2023. Generative artificial intelligence: can ChatGPT write a quality abstract? *Emergency Medicine Australasia* 35, 809–811.

Bana e Costa, C.A., Vansnick, J.-C., 1994. MACBETH—an interactive path towards the construction of cardinal value functions. *International Transactions in Operational Research* 1, 489–500.

Bolaños, F., Salatino, A., Osborne, F., Motta, E., 2024. Artificial intelligence for literature reviews: opportunities and challenges review. *Artificial Intelligence Review* 57, 259.

Borah, R., Brown, A.W., Capers, P.L., Kaiser, K.A., 2017. Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open* 7, e012545.

Bornmann, L., Lepori, B., 2024. The use of ChatGPT to find similar institutions for institutional benchmarking. *Scientometrics* 129, 3593–3598.

Brandtzaeg, P.B., Følstad, A., 2017. Why people use chatbots. In Kompatsiaris, I., Cave, J., Satsiou, A., Carle, G., Passani, A., Kontopoulos, E., Diplaris, S., McMillan, D. (eds) *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 10673 LNCS*, Springer, Cham, pp. 377–392.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., 2020. Language models are few-shot learners. Advances in Neural Information Processing Systems 33, November 28–December 9, New Orleans, LA, pp. 1877–1901.

Chaudhry, I.A., Khan, A.A., 2016. A research survey: review of flexible job shop scheduling techniques. *International Transactions in Operational Research* 23, 551–591.

Chelli, M., Descamps, J., Lavoué, V., Trojani, C., Azar, M., Deckert, M., Raynier, J.L., Clowez, G., Boileau, P., Ruetsch-Chelli, C., 2024. Hallucination rates and reference accuracy of ChatGPT and Bard for systematic reviews: comparative analysis. *Journal of Medical Internet Research* 26, e53164.

Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A.N., Li, T., Li, D., Zhu, B., Zhang, H., Jordan, M.I., Gonzalez, J.E., 2024. Chatbot arena: an open platform for evaluating LLMs by human preference. arXiv preprint. Available at https://arxiv.org/abs/2403.04132 (accessed 28 August 2024).

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Colorni, A., Dorigo, M., Maffioli, F., Maniezzo, V., Righini, G., Trubian, M., 1996. Heuristics from nature for hard combinatorial optimization problems. *International Transactions in Operational Research* 3, 1–21.

Constantin, I., Florian, M., 1995. Optimizing frequencies in a transit network: a nonlinear bi-level programming approach. *International Transactions in Operational Research* 2, 149–164.

Dabbagh, N., 2001. Concept mapping as a mindtool for critical thinking. *Journal of Computing in Teacher Education* 17, 16–23.

de Freitas, J.C., Penna, P.H. V, 2020. A variable neighborhood search for flying sidekick traveling salesman problem. *International Transactions in Operational Research* 27, 267–290.

de Winter, J., 2024. Can ChatGPT be used to predict citation counts, readership, and social media interaction? An exploration among 2222 scientific abstracts. *Scientometrics* 129, 2469–2487.

Delacroix, S., 2023. Sustainable data rivers? SSRN Electronic Journal.

Dergaa, I., Chamari, K., Zmijewski, P., Saad, H. Ben, 2023. From human writing to artificial intelligence generated text: examining the prospects and potential threats of ChatGPT in academic writing. *Biology of Sport* 40, 615.

Dwivedi, Y.K., Kshetri, N., Hughes, L., Slade, E.L., Jeyaraj, A., Kar, A.K., Baabdullah, A.M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M.A., Al-Busaidi, A.S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., Carter, L., Chowdhury, S., Crick, T., Cunningham, S.W., Davies, G.H., Davison, R.M., Dé, R., Dennehy, D., Duan, Y., Dubey, R., Dwivedi, R., Edwards, J.S., Flavián, C., Gauld, R., Grover, V., Hu, M.C., Janssen, M., Jones, P., Junglas, I., Khorana, S., Kraus, S., Larsen, K.R., Latreille, P., Laumer, S., Malik, F.T., Mardani, A., Mariani, M., Mithas, S., Mogaji, E., Nord, J.H., O'Connor, S., Okumus, F., Pagani, M., Pandey, N., Papagiannidis, S., Pappas, I.O., Pathak, N., Pries-Heje, J., Raman, R., Rana, N.P., Rehm, S.V., Ribeiro-Navarrete, S., Richter, A., Rowe, F., Sarker, S., Stahl, B.C., Tiwari, M.K., van der Aalst, W., Venkatesh, V., Viglia, G., Wade, M., Walton, P., Wirtz, J., Wright, R., 2023. Opinion paper: "so what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management* 71, 102642.

European Parliament, 2023. EU AI Act: first regulation on artificial intelligence. Available at https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence (accessed 28 August 2024).

Fagerholt, K., 1999. Optimal fleet design in a ship routing problem. *International Transactions in Operational Research* 6, 453–464.

Farhat, F., Sohail, S.S., Madsen, D.Ø., 2023. How trustworthy is ChatGPT? The case of bibliometric analyses. *Cogent Engineering* 10, 1, 2222988.

Festa, P., Resende, M.G.C., 2009a. An annotated bibliography of GRASP—part I: algorithms. *International Transactions in Operational Research* 16, 1–24.

Festa, P., Resende, M.G.C., 2009b. An annotated bibliography of GRASP—part II: applications. *International Transactions in Operational Research* 16, 131–172.

Feuerriegel, S., Hartmann, J., Janiesch, C., Zschech, P., 2023. *Generative ai. Business & Information Systems Engineering*, 66(1), 111–126.

Gao, C.A., Howard, F.M., Markov, N.S., Dyer, E.C., Ramesh, S., Luo, Y., Pearson, A.T., 2023. Comparing scientific abstracts generated by ChatGPT to real abstracts with detectors and blinded human reviewers. *npj Digital Medicine* 6, 1–5.

Gehring, H., Bortfeldt, A., 1997. A genetic algorithm for solving the container loading problem. *International Transactions in Operational Research* 4, 401–418.

Gehring, H., Bortfeldt, A. 2002. A parallel genetic algorithm for solving the container loading problem. *International Transactions in Operational Research*, 9(4), 497–511.

Glikson, E., Woolley, A.W., 2020. Human trust in artificial intelligence: review of empirical research. *Academy of Management Annals* 14, 627–660.

Google., 2023. Gemini. Available at https://gemini.google.com/app (accessed c).

Governo Italiano., 2021. Programma Strategico Intelligenza Artificiale 2022–2024. Available at: https://assets.innovazione.gov.it/1637777289-programma-strategico-iaweb.pdf (accessed 28 August 2024).

Guajardo, M., Rönnqvist, M., 2016. A review on cost allocation methods in collaborative transportation. *International Transactions in Operational Research* 23, 371–392.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Guimarães, N.S., Joviano-Santos, J.V., Reis, M.G., Chaves, R.R.M., 2024. Development of search strategies for systematic reviews in health using ChatGPT: a critical analysis. *Journal of Translational Medicine* 22, 1–3.

Hagendorff, T., 2024. Mapping the ethics of generative AI: A comprehensive scoping review. *Minds and Medicine* 34, 39.

Haman, M., Školník, M., 2024. Using ChatGPT to conduct a literature review. *Accountability in Research* 31, 8, 1244–1246.

Han, X., Zhang, Z., Ding, N., Gu, Y., Liu, X., Huo, Y., Qiu, J., Yao, Y., Zhang, A., Zhang, L., Han, W., Huang, M., Jin, Q., Lan, Y., Liu, Y., Liu, Z., Lu, Z., Qiu, X., Song, R., Tang, J., Wen, J.R., Yuan, J., Zhao, W.X., Zhu, J., 2021. Pre-trained models: past, present and future. *AI Open* 2, 225–250.

Hicks, M.T., Humphries, J., Slater, J., 2024. ChatGPT is bullshit. *Ethics and Information Technology* 26, 1–10.

Hoogkamer, 2024. Termboard. Termboard. Available at https://termboard.com/app#/ (accessed 28 August 2024).

Hosseini, M., Rasmussen, L.M., Resnik, D.B., 2023. Using AI to write scholarly publications. *Accountability in Research* 31, 7, 715–723.

Ioannidis, J.P.A., Fanelli, D., Dunne, D.D., Goodman, S.N., 2015. Meta-research: evaluation and improvement of research methods and practices. *PLoS Biology* 13, e1002264.

ITOR, 2024. Overview—International Transactions in Operational Research. Available at https://onlinelibrary.wiley.com/page/journal/14753995/homepage/productinformation.html (accessed 28 August 2024).

Jonassen, D.H., 1996. *Computers in the Classroom: Mindtools for Critical Thinking*. Prentice-Hall, Inc., Englewood Cliffs, NJ.

Keiper, M.C., 2023. ChatGPT in practice: increasing event planning efficiency through artificial intelligence. *Journal of Hospitality, Leisure, Sport & Tourism Education* 33, 100454.

Klenk, M., 2024. Ethics of generative AI and manipulation: a design-oriented research agenda. *Ethics and Information Technology* 26, 1–15.

Kozan, E., Preston, P., 1999. Genetic algorithms to schedule container transfers at multimodal terminals. *International Transactions in Operational Research* 6, 311–329.

Kumar, B., Kumar, T., Nadakuditi, S., 2023. Ethical implications of Generative AI: the case of large language models. *Journal of Scientific and Engineering Research* 10, 122–127.

Laporte, G., Gendreau, M., Potvin, J.-Y., Semet, F., 2000. Classical and modern heuristics for the vehicle routing problem. *International Transactions in Operational Research* 7, 285–300.

Liberti, L., Lavor, C., Maculan, N., 2008. A Branch-and-prune algorithm for the molecular distance geometry problem. *International Transactions in Operational Research* 15, 1–17.

Lipton, Z.C., 2018. The mythos of model interpretability. *Communications of the ACM* 61, 35–43.

Malaguti, E., Toth, P. 2010. A survey on vertex coloring problems. *International transactions in operational research*, 17(1), 1–34.

Marquis, Y., Oladoyinbo, T. O., Olabanji, S. O., Olaniyi, O. O., Ajayi, S. A. 2024. Proliferation of AI tools: A multifaceted evaluation of user perceptions and emerging trend. *Asian Journal of Advanced Research and Reports*, 18(1), 30–55.

Marzuki, Widiati, U., Rusdin, D., Darwin, Indrawati, I., 2023. The impact of AI writing tools on the content and organization of students' writing: EFL teachers' perspective. *Cogent Education* 10, 2236469.

Meyer, J.G., Urbanowicz, R.J., Martin, P.C.N., O'Connor, K., Li, R., Peng, P.C., Bright, T.J., Tatonetti, N., Won, K.J., Gonzalez-Hernandez, G., Moore, J.H., 2023. ChatGPT and large language models in academia: opportunities and challenges. *BioData Mining* 16, 1–11.

Microsoft, 2023. CoPilot. Available at https://copilot.microsoft.com/?culture=en-us (accessed 28 Augut 2024).

Mingers, J., 2000. Variety is the spice of life: combining soft and hard OR/MS methods. *International Transactions in Operational Research* 7, 673–691.

Miranda, B., Lee, A., Sundar, S., Casasola, A., Koyejo, S. 2023. Beyond Scale: The Diversity Coefficient as a Data Quality Metric for Variability in Natural Language Data. arXiv preprint arXiv:2306.13840. Available at https://arxiv.org/abs/2306.13840 (accessed 28 August 2024).

Mistral., 2024. LeChat. Available at https://chat.mistral.ai/chat (accessed 28 August 2024).

Narbón-Perpiñá, I., De Witte, K., 2018a. Local governments' efficiency: a systematic literature review—part I. *International Transactions in Operational Research* 25, 431–468.

Narbón-Perpiñá, I., De Witte, K., 2018b. Local governments' efficiency: a systematic literature review—part II. *International Transactions in Operational Research* 25, 1107–1136.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

O'Neill, M.M., Booth, S.R., Lamb, J.T., 2018. Using NVivo™ for literature reviews: the eight step pedagogy (N7+ 1). *Qualitative Report* 23, 21–39.

OpenAI, 2022. ChatGPT. ChatGPT. Availabl at https://chatgpt.com/(accessed 28 August 2024).

Osman, I.H., Christofides, N., 1994. Capacitated clustering problems by hybrid simulated annealing and tabu search. *International Transactions in Operational Research* 1, 317–336.

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372, n71.

Pérez-Bernabeu, E., Juan, A. A., Faulin, J., Barrios, B. B. 2015. Horizontal cooperation in road transportation: A case illustrating savings in distances and greenhouse gas emissions. *International Transactions in Operational Research*, 22(3), 585–606.

Perplexity AI, Inc., 2022. Perplexity. Available at https://www.perplexity.ai/ (accessed 28 August 2024).

Petticrew, M., Roberts, H., 2008. *Systematic Reviews in the Social Sciences: A Practical Guide*. John Wiley & Sons, Oxford.

Preux, P., Talbi, E. G. 1999. Towards hybrid evolutionary algorithms. *International transactions in operational research*, 6(6), 557–570.

Radford, A., Wu, J., Amodei, Dario, Amodei, Daniela, Clark, J., Brundage, M., Sutskever, I., 2019. Better language models and their implications. OpenAI blog. Available at https://openai.com/index/better-language-models/ (accessed 28 August 2024).

Raiaan, M.A.K., Mukta, M.S.H., Fatema, K., Fahad, N.M., Sakib, S., Mim, M.M.J., Ahmad, J., Ali, M.E., Azam, S., 2024. A review on large language models: architectures, applications, taxonomies, open issues and challenges. *IEEE Access* 12, 26839–26874.

Rais, A., Viana, A., 2011. Operations research in healthcare: a survey. *International Transactions in Operational Research* 18, 1–31.

Rojas Viloria, D., Solano-Charris, E.L., Muñoz-Villamizar, A., Montoya-Torres, J.R., 2021. Unmanned aerial vehicles/drones in vehicle routing problems: a literature review. *International Transactions in Operational Research* 28, 1626–1657.

Schank, R.C., 1987. What is AI, anyway? *AI Magazine* 8, 59–59.

Schank, R.C., 1991. Where's the AI? *AI Magazine* 12, 38–38.

Scott, T. J., Read, E. G. 1996. Modelling hydro reservoir operation in a deregulated electricity market. *International Transactions in Operational Research*, 3(3–4), 243–253.

Sheikh, H., Prins, C., Schrijvers, E., 2023. Artificial intelligence: definition and background. In Sheikh, H., Prins, C., Schrijvers, E. (eds) *Mission AI. Research for Policy*. Springer, Cham, pp. 15–41.

Shen, Y., Heacock, L., Elias, J., Hentel, K.D., Reig, B., Shih, G., Moy, L., 2023. ChatGPT and other large language models are double-edged swords. *Radiology* 307, 2023.

Shojania, K.G., Sampson, M., Ansari, M.T., Ji, J., Doucette, S., Moher, D., 2007. How quickly do systematic reviews go out of date? A survival analysis. *Annals of Internal Medicine* 147, 224–233.

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., Gal, Y., 2024. AI models collapse when trained on recursively generated data. *Nature* 631, 755–759.

Sinuany-Stern, Z., Mehrez, A., Hadad, Y., 2000. An AHP/DEA methodology for ranking decision making units. *International Transactions in Operational Research* 7, 109–124.

Sörensen, K., 2015. Metaheuristics—the metaphor exposed. *International Transactions in Operational Research* 22, 3–18.

Syriani, E., David, I., Kumar, G., 2024. Screening articles for systematic reviews with ChatGPT. *Journal of Computer Languages* 80, 101287.

Tang, L., Zhou, X., Lu, M., 2024. An ensemble approach for research article identification: a case study in artificial intelligence. *PeerJ Computer Science* 10, e2521.

Tomasini, R.M., Van Wassenhove, L.N., 2009. From preparedness to partnerships: case study research on humanitarian logistics. *International Transactions in Operational Research* 16, 549–559.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Torres Salinas, D., Thelwall, M., Arroyo Machado, W., 2025. ChatGPT for bibliometrics: potential applications and limitations. In Gooding, P., Terras, M.M., Ames, S. (eds) Library Catalogues as Data: Research, Practice, and Usage. Facet Publishing, London.

Turing, A.M., 2009. *Computing Machinery and Intelligence*. Springer, Cham.

Van der Vorst, J.G.A.J., Beulens, A.J.M., De Wit, W., Van Beek, P., 1998. Supply chain management in food chains: improving performance by reducing uncertainty. *International Transactions in Operational Research* 5, 487–499.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., et al. 2017. Attention is all you need. Advances in neural information processing systems, 30. Available at https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html (accessed 28 August 2024).

Vatican State, 2020. Rome call for AI ethics. Available at https://www.vatican.va/roman_curia/pontifical_academies/acdlife/documents/rc_pont-acd_life_doc_20202228_rome-call-for-ai-ethics_en.pdf (accessed 28 August 2024).

Vu, K.K., D'Ambrosio, C., Hamadi, Y., Liberti, L., 2017. Surrogate-based methods for black-box optimization. *International Transactions in Operational Research* 24, 393–424.

Walter, Y., 2024. Embracing the future of Artificial Intelligence in the classroom: the relevance of AI literacy, prompt engineering, and critical thinking in modern education. *International Journal of Educational Technology in Higher Education* 21, 1–29.

Wang, S., Scells, H., Koopman, B., Zuccon, G., 2023. Can ChatGPT write a good Boolean query for systematic review literature search? SIGIR 2023—Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, July 23–27, Taipei, Taiwan, pp. 1426–1436.

Van Wassenhove, L.N., Pedraza Martinez, A.J., 2012. Using OR to adapt supply chain management best practices to humanitarian logistics. *International Transactions in Operational Research* 19, 307–322.

Wei, J., Bosma, M., Zhao, V.Y., Guu, K., Yu, A.W., Lester, B., Du, N., Dai, A.M., Le, Q. V, 2021. Finetuned language models are zero-shot learners. arXiv preprint arXiv:2109.01652. Available at https://arxiv.org/abs/2109.01652 (accessed 22 January 2025).

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V, Zhou, D., 2022. Chain-of-thought prompting elicits reasoning in large language models. Advances in Neural Information Processing Systems 35, November 28–December 9, New Orleans, LA, pp. 24824–24837.

Yeh, C.-H., 2002. A problem-based selection of multi-attribute decision-making methods. *International Transactions in Operational Research* 9, 169–181.

Yu, S.-M., Wang, J., Wang, J.-Q., 2018. An extended TODIM approach with intuitionistic linguistic numbers. *International Transactions in Operational Research* 25, 781–805.

Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y., 2019. Bertscore: Evaluating text generation with bert. arXiv preprint arXiv:1904.09675. Available at https://arxiv.org/abs/1904.09675 (accessed 22 August 2024).

Appendix A

Table A1
Search results by CoPilot and ChatGPT. Date of retrieval: August 2, 2024

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
500+	10.1111/itor.12001	Glover, F. (2013). Metaheuristics—the metaphor exposed. *International Transactions in Operational Research*, 20(1), 3–18.	CoPilot (GPT-4)
450+	10.1111/itor.12002	Zhang, R., & Cheng, T. C. E. (2013). A research survey: review of flexible job shop scheduling techniques. *International Transactions in Operational Research*, 20(2), 309–329.	CoPilot (GPT-4)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1
(Continued)

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
400+	10.1111/itor.12003	Dorling, K., Heinrichs, J., Messier, G. G., & Magierowski, S. (2013). Unmanned aerial vehicles/drones in vehicle routing problems: a literature review. *International Transactions in Operational Research*, 20(3), 353–374.	CoPilot (GPT-4)
350+	10.1111/itor.12004	Hansen, P., & Mladenović, N. (2013). A variable neighborhood search for flying sidekick traveling salesman problem. *International Transactions in Operational Research*, 20(4), 475–495.	CoPilot (GPT-4)
300+	10.1111/itor.12005	Guajardo, M., & Rönnqvist, M. (2013). A review on cost allocation methods in collaborative transportation. *International Transactions in Operational Research*, 20(5), 675–695.	CoPilot (GPT-4)
250+	10.1111/itor.12006	Kovács, G., & Spens, K. M. (2013). From preparedness to partnerships: case study research on humanitarian logistics. *International Transactions in Operational Research*, 20(6), 1027–1047.	CoPilot (GPT-4)
200+	10.1111/itor.12007	Crainic, T. G., & Toulouse, M. (2013). Parallel metaheuristics: recent advances and new trends. *International Transactions in Operational Research*, 20(7), 1234–1254.	CoPilot (GPT-4)
180+	10.1111/itor.12008	Worthington, A. C., & Dollery, B. E. (2013). Local governments' efficiency: a systematic literature review—part I. *International Transactions in Operational Research*, 20(8), 1456–1476.	CoPilot (GPT-4)
160+	10.1111/itor.12009	Rais, A., & Viana, A. (2013). Operations Research in Healthcare: a survey. *International Transactions in Operational Research*, 20(9), 1678–1698.	CoPilot (GPT-4)
150+	10.1111/itor.12010	Keskin, M., & Çatay, B. (2013). The electric vehicle routing problem with shared charging stations. *International Transactions in Operational Research*, 20(10), 1890–1910.	CoPilot (GPT-4)
140+	10.1111/itor.12011	Taleizadeh, A. A., & Niaki, S. T. A. (2013). Price discount facility in an EOQ model for deteriorating items with stock-dependent demand and partial backlogging. *International Transactions in Operational Research*, 20(11), 2112–2132.	CoPilot (GPT-4)
130+	10.1111/itor.12012	Zavadskas, E. K., & Turskis, Z. (2013). Techniques to model uncertain input data of multi-criteria decision-making problems: a literature review. *International Transactions in Operational Research*, 20(12), 2334–2354.	CoPilot (GPT-4)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1
(Continued)

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
120+	10.1111/itor.12013	Worthington, A. C., & Dollery, B. E. (2013). Local governments' efficiency: a systematic literature review—part II. *International Transactions in Operational Research*, 20(13), 2556–2576.	CoPilot (GPT-4)
110+	10.1111/itor.12014	Cook, W. D., & Zhu, J. (2013). A review of Dynamic Data Envelopment Analysis: state of the art and applications. *International Transactions in Operational Research*, 20(14), 2778–2798.	CoPilot (GPT-4)
100+	10.1111/itor.12015	Cruijssen, F., & Salomon, M. (2013). A survey on obstacles and difficulties of practical implementation of horizontal collaboration in logistics. *International Transactions in Operational Research*, 20(15), 2990–3010.	CoPilot (GPT-4)
90+	10.1111/itor.12016	Bortfeldt, A., & Wäscher, G. (2013). A comparative review of 3D container loading algorithms. *International Transactions in Operational Research*, 20(16), 3212–3232.	CoPilot (GPT-4)
80+	10.1111/itor.12017	Jones, D. R., & Schonlau, M. (2013). Surrogate-based methods for black-box optimization. *International Transactions in Operational Research*, 20(17), 3434–3454.	CoPilot (GPT-4)
70+	10.1111/itor.12018	Rockafellar, R. T., & Uryasev, S. (2013). Conditional value-at-risk beyond finance: a survey. *International Transactions in Operational Research*, 20(18), 3656–3676.	CoPilot (GPT-4)
60+	10.1111/itor.12019	Yang, Z., & Zhang, H. (2013). Financing strategies for a capital-constrained manufacturer in a dual-channel supply chain. *International Transactions in Operational Research*, 20(19), 3878–3898.	CoPilot (GPT-4)
50+	10.1111/itor.12020	Li, Y., & Liu, J. (2013). Financing decisions in supply chains with a capital-constrained manufacturer: competition and risk. *International Transactions in Operational Research*, 20(20), 4090–4110.	CoPilot (GPT-4)
45+	10.1111/itor.12021	Bertsimas, D., & Sim, M. (2013). Robust discrete optimization and network flows. *International Transactions in Operational Research*, 20(21), 4312–4332.	CoPilot (GPT-4)
40+	10.1111/itor.12022	Laporte, G., & Nobert, Y. (2013). Exact algorithms for thae vehicle routing problem. *International Transactions in Operational Research*, 20(22), 4534–4554.	CoPilot (GPT-4)
35+	10.1111/itor.12023	Pisinger, D., & Ropke, S. (2013). Large neighborhood search. *International Transactions in Operational Research*, 20(23), 4756–4776.	CoPilot (GPT-4)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1
(Continued)

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
30+	10.1111/itor.12024	Toth, P., & Vigo, D. (2013). Models, relaxations and exact approaches for the capacitated vehicle routing problem. *International Transactions in Operational Research*, 20(24), 4978–4998.	CoPilot (GPT-4)
25+	10.1111/itor.12025	Golden, B., & Wasil, E. (2013). Metaheuristics in combinatorial optimization: overview and conceptual comparison. *International Transactions in Operational Research*, 20(25), 5200–5220.	CoPilot (GPT-4)
20+	10.1111/itor.12026	Clarke, G., & Wright, J. W. (2013). Scheduling of vehicles from a central depot to a number of delivery points. *International Transactions in Operational Research*, 20(26), 5422–5442.	CoPilot (GPT-4)
15+	10.1111/itor.12027	Fisher, M. L. (2013). Optimal solution of vehicle routing problems using minimum k-trees. *International Transactions in Operational Research*, 20(27), 5644–5664.	CoPilot (GPT-4)
10+	10.1111/itor.12028	Gendreau, M., & Potvin, J. Y. (2013). Dynamic vehicle routing and dispatching. *International Transactions in Operational Research*, 20(28), 5866–5886.	CoPilot (GPT-4)
5+	10.1111/itor.12029	Laporte, G., & Osman, I. H. (2013). Routing problems: a bibliography. *International Transactions in Operational Research*, 20(29), 6088–6108.	CoPilot (GPT-4)
5+	10.1111/itor.12030	Bramel, J., & Simchi-Levi, D. (2013). The logic of logistics: theory, algorithms, and applications for logistics management. *International Transactions in Operational Research*, 20(30), 6310–6330.	CoPilot (GPT-4)
650+	10.1111/j.1475-3995.1987.tb00409.x	Kusiak, A., & Heragu, S. S. (1987). The facility layout problem. *International Transactions in Operational Research*, *12*(4), 381–396. https://doi.org/10.1111/j.1475-3995.1987.tb00409.x	ChatGPT (GPT-4o)
620+	10.1111/j.1475-3995.2003.00529.x	Bertsimas, D., & Sim, M. (2003). The price of robustness. *International Transactions in Operational Research*, *50*(1), 66–95. https://doi.org/10.1111/j.1475-3995.2003.00529.x	ChatGPT (GPT-4o)
590+	10.1111/j.1475-3995.1996.tb00724.x	Bienstock, D. (1996). Computational study of a family of mixed-integer quadratic programming problems. *International Transactions in Operational Research*, *45*(2), 287–307. https://doi.org/10.1111/j.1475-3995.1996.tb00724.x	ChatGPT (GPT-4o)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1
(Continued)

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
550+	10.1111/j.1475-3995.1997.tb00458.x	Bramel, J., & Simchi-Levi, D. (1997). The logic of logistics: Theory, algorithms, and applications for logistics management. *International Transactions in Operational Research*, *23*(4), 110–126. https://doi.org/10.1111/j.1475-3995.1997.tb00458.x	ChatGPT (GPT-4o)
500+	10.1111/j.1475-3995.1994.tb00142.x	Kall, P., & Wallace, S. W. (1994). Stochastic programming. *International Transactions in Operational Research*, *21*(1), 7–19. https://doi.org/10.1111/j.1475-3995.1994.tb00142.x	ChatGPT (GPT-4o)
470+	10.1111/j.1475-3995.1997.tb00718.x	Aarts, E., & Lenstra, J. K. (1997). Local search in combinatorial optimization. *International Transactions in Operational Research*, *45*(3), 77–105. https://doi.org/10.1111/j.1475-3995.1997.tb00718.x	ChatGPT (GPT-4o)
460+	10.1111/j.1475-3995.1995.tb00186.x	Smith, R. L., & Pantelides, C. C. (1995). Simple bounds for uncertain variables in multiperiod optimization. *International Transactions in Operational Research*, *13*(2), 202–215. https://doi.org/10.1111/j.1475-3995.1995.tb00186.x	ChatGPT (GPT-4o)
440+	10.1111/j.1475-3995.2001.tb00517.x	Günlük, O., & Pochet, Y. (2001). Polyhedral study of the knapsack problem with generalized upper bounds. *International Transactions in Operational Research*, *37*(3), 485–500. https://doi.org/10.1111/j.1475-3995.2001.tb00517.x	ChatGPT (GPT-4o)
430+	10.1111/j.1475-3995.1990.tb00119.x	Bazaraa, M. S., & Sherali, H. D. (1990). A study of a class of exact penalty functions for nonlinear programming. *International Transactions in Operational Research*, *17*(4), 143–157. https://doi.org/10.1111/j.1475-3995.1990.tb00119.x	ChatGPT (GPT-4o)
420+	10.1111/j.1475-3995.1997.tb00723.x	De Boer, R., & Klundert, J. (1997). Solving large scale facility location problems: A case study from the food industry. *International Transactions in Operational Research*, *44*(2), 215–227. https://doi.org/10.1111/j.1475-3995.1997.tb00723.x	ChatGPT (GPT-4o)
410+	10.1111/j.1475-3995.1999.tb00487.x	Nemhauser, G. L., & Wolsey, L. A. (1999). Integer and combinatorial optimization. *International Transactions in Operational Research*, *32*(3), 127–142. https://doi.org/10.1111/j.1475-3995.1999.tb00487.x	ChatGPT (GPT-4o)
400+	10.1111/j.1475-3995.1998.tb00503.x	Pardalos, P. M., & Resende, M. G. (1998). Handbook of applied optimization. *International Transactions in Operational Research*, *39*(4), 243–256. https://doi.org/10.1111/j.1475-3995.1998.tb00503.x	ChatGPT (GPT-4o)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1
(Continued)

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
390+	10.1111/j.1475-3995.2000.tb00415.x	Hooker, J. N. (2000). Logic-based methods for optimization: Combining optimization and constraint satisfaction. *International Transactions in Operational Research*, *41*(2), 77–92. https://doi.org/10.1111/j.1475-3995.2000.tb00415.x	ChatGPT (GPT-4o)
380+	10.1111/j.1475-3995.1999.tb00507.x	Ben-Tal, A., & Nemirovski, A. (1999). Robust solutions of uncertain linear programs. *International Transactions in Operational Research*, *48*(3), 297–309. https://doi.org/10.1111/j.1475-3995.1999.tb00507.x	ChatGPT (GPT-4o)
370+	10.1111/j.1475-3995.2002.tb00517.x	Mitra, G., & Vazacopoulos, A. (2002). Decision support systems: Tools and techniques for interactive decision making. *International Transactions in Operational Research*, *47*(1), 199–216. https://doi.org/10.1111/j.1475-3995.2002.tb00517.x	ChatGPT (GPT-4o)
360+	10.1111/j.1475-3995.1995.tb00479.x	Gal, T., & Greenberg, H. J. (1995). Advances in sensitivity analysis and parametric programming. *International Transactions in Operational Research*, *29*(2), 97–112. https://doi.org/10.1111/j.1475-3995.1995.tb00479.x	ChatGPT (GPT-4o)
350+	10.1111/j.1475-3995.2001.tb00412.x	Laporte, G., & Osman, I. H. (2001). Metaheuristics in combinatorial optimization. *International Transactions in Operational Research*, *35*(1), 66–89. https://doi.org/10.1111/j.1475-3995.2001.tb00412.x	ChatGPT (GPT-4o)
340+	10.1111/j.1475-3995.1997.tb00523.x	Birge, J. R., & Louveaux, F. (1997). Introduction to stochastic programming. *International Transactions in Operational Research*, *32*(4), 45–60. https://doi.org/10.1111/j.1475-3995.1997.tb00523.x	ChatGPT (GPT-4o)
330+	10.1111/j.1475-3995.1995.tb00578.x	Fisher, M. L. (1995). Vehicle routing. *International Transactions in Operational Research*, *27*(3), 207–221. https://doi.org/10.1111/j.1475-3995.1995.tb00578.x	ChatGPT (GPT-4o)
320+	10.1111/j.1475-3995.1993.tb00518.x	Nemhauser, G. L., & Wolsey, L. A. (1993). Integer programming. *International Transactions in Operational Research*, *30*(2), 127–149. https://doi.org/10.1111/j.1475-3995.1993.tb00518.x	ChatGPT (GPT-4o)
310+	10.1111/j.1475-3995.1999.tb00513.x	Horst, R., & Pardalos, P. M. (1999). Handbook of global optimization. *International Transactions in Operational Research*, *42*(3), 311–327. https://doi.org/10.1111/j.1475-3995.1999.tb00513.x	ChatGPT (GPT-4o)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A1
(Continued)

Number of citations	Digital object identifier (DOI)	APA reference	ChatBot
300+	10.1111/j.1475-3995.2004.tb00509.x	Ruszczynski, A. (2004). Decomposition methods in stochastic programming. *International Transactions in Operational Research*, *48*(1), 21–33. https://doi.org/10.1111/j.1475-3995.2004.tb00509.x	ChatGPT (GPT-4o)
290+	10.1111/j.1475-3995.2001.tb00514.x	Hooker, J. N. (2001). Integrated methods for optimization. *International Transactions in Operational Research*, *35*(2), 104–118. https://doi.org/10.1111/j.1475-3995.2001.tb00514.x	ChatGPT (GPT-4o)
280+	10.1111/j.1475-3995.1997.tb00457.x	Crainic, T. G., & Laporte, G. (1997). Planning models for freight transportation. *International Transactions in Operational Research*, *20*(4), 153–171. https://doi.org/10.1111/j.1475-3995.1997.tb00457.x	ChatGPT (GPT-4o)
270+	10.1111/j.1475-3995.2000.tb00406.x	Dantzig, G. B., & Thapa, M. N. (2000). Linear programming: Foundations and extensions. *International Transactions in Operational Research*, *36*(1), 1–18. https://doi.org/10.1111/j.1475-3995.2000.tb00406.x	ChatGPT (GPT-4o)
260+	10.1111/j.1475-3995.1999.tb00510.x	Smith, J. E. (1999). Dynamic programming and the control of processes. *International Transactions in Operational Research*, *48*(2), 213–226. https://doi.org/10.1111/j.1475-3995.1999.tb00510.x	ChatGPT (GPT-4o)
250+	10.1111/j.1475-3995.1996.tb00431.x	Nocedal, J., & Wright, S. J. (1996). Numerical optimization. *International Transactions in Operational Research*, *27*(3), 197–212. https://doi.org/10.1111/j.1475-3995.1996.tb00431.x	ChatGPT (GPT-4o)
240+	10.1111/j.1475-3995.2002.tb00479.x	Chinneck, J. W. (2002). Feasibility and infeasibility in optimization: Algorithms and computational methods. *International Transactions in Operational Research*, *34*(4), 287–303. https://doi.org/10.1111/j.1475-3995.2002.tb00479.x	ChatGPT (GPT-4o)
230+	10.1111/j.1475-3995.1997.tb00503.x	Fulkerson, D. R., & Harding, G. H. (1997). Maximizing the minimum source-sink path length. *International Transactions in Operational Research*, *32*(2), 47–61. https://doi.org/10.1111/j.1475-3995.1997.tb00503.x	ChatGPT (GPT-4o)
220+	10.1111/j.1475-3995.1998.tb00479.x	Kennington, J. L., & Helgason, R. V. (1998). Algorithms for network programming. *International Transactions in Operational Research*, *35*(4), 165–178. https://doi.org/10.1111/j.1475-3995.1998.tb00479.x	ChatGPT (GPT-4o)

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix B

Table B1
Data extraction results by Claude, CoPilot and ChatGPT. Date of retrieval: August 19, 2024

Article	AI-ChatBot (model)	Topic extracted
Sörensen (2015)	Claude 3.5 Sonnet	Critiques metaphor-based metaheuristics
Laporte et al. (2000)	Claude 3.5 Sonnet	Over the limit
Bana e Costa and Vansnick (1994)	Claude 3.5 Sonnet	Over the limit
Chaudhry and Khan (2016)	Claude 3.5 Sonnet	Over the limit
Rais and Viana (2011)	Claude 3.5 Sonnet	Over the limit
Tomasini and Van Wassenhove (2009)	Claude 3.5 Sonnet	Evolution of SCM in disaster relief
Alba et al. (2013)	Claude 3.5 Sonnet	Over the limit
Sinuany-Stern et al. (2000)	Claude 3.5 Sonnet	A hybrid analytical hierarchical process (AHP)/data envelopment analysis (DEA) methodology for ranking decision making
Colorni et al. (1996)	Claude 3.5 Sonnet	Not able to access the content
Gehring and Bortfeldt (1997)	Claude 3.5 Sonnet	Not able to access the content
Guajardo and Rönnqvist (2016)	Claude 3.5 Sonnet	Over the limit
Osman and Christofides (1994)	Claude 3.5 Sonnet	Over the imit
Fagerholt (1999)	Claude 3.5 Sonnet	method for determining the optimal fleet size and routing for a liner shipping problem
Malaguti and Toth (2010)	Claude 3.5 Sonnet	Over the limit
Archetti and Speranza (2012)	Claude 3.5 Sonnet	Over the limit
Van Wassenhove and Pedraza Martinez (2012)	Claude 3.5 Sonnet	The paper demonstrates the potential for adapting proven SCM practices to the humanitarian sector using operations research (OR) methods while highlighting the importance of understanding the unique context
Rojas Viloria et al. (2021)	Claude 3.5 Sonnet	Over the limit
Festa and Resende (2009a)	Claude 3.5 Sonnet	Over the limit
Kozan and Preston (1999)	Claude 3.5 Sonnet	Application of GAs to optimize container transfer scheduling and storage strategies at multimodal seaport terminals, with the goal of minimizing ship turnaround times
Yu et al. (2018)	Claude 3.5 Sonnet	Over the limit
Narbón-Perpiñá and De Witte (2018a)	Claude 3.5 Sonnet	Over the limit
Festa and Resende (2009b)	Claude 3.5 Sonnet	Over the limit
Van der Vorst et al. (1998)	Claude 3.5 Sonnet	SCM strategies to reduce uncertainty and improve performance in food supply chains, focusing on a case study of chilled salads
de Freitas and Penna (2020)	Claude 3.5 Sonnet	Over the limit
Constantin and Florian (1995)	Claude 3.5 Sonnet	Not able to access the content

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B1
(Continued)

Article	AI-ChatBot (model)	Topic extracted
Yeh (2002)	Claude 3.5 Sonnet	A new approach for validating and selecting MADM methods based on sensitivity analysis of attribute weights and Shannon's entropy concept
Narbón-Perpiñá and De Witte (2018b)	Claude 3.5 Sonnet	Over the limit
Mingers (2000)	Claude 3.5 Sonnet	Benefits of combining multiple "soft" and "hard" operational research/management science (OR/MS) methods
Vu et al. (2017)	Claude 3.5 Sonnet	Over the limit
Liberti et al. (2008)	Claude 3.5 Sonnet	The main topic of this paper is a branch-and-prune algorithm for solving the molecular distance geometry problem (MDGP), which aims to determine the 3D structure of molecules given some inter-atomic distances
Sörensen (2015)	ChatGPT (GPT-4o)	The document criticizes the proliferation of new metaheuristic algorithms based on superficial metaphors
Laporte et al. (2000)	ChatGPT (GPT-4o)	A comprehensive survey of classical and modern heuristics for the VRP
Bana e Costa and Vansnick (1994)	ChatGPT (GPT-4o)	MACBETH approach, a method for constructing cardinal value functions through an interactive process using pairwise verbal judgments of attractiveness differences
Chaudhry and Khan (2016)	ChatGPT (GPT-4o)	A comprehensive survey of flexible job shop scheduling (FJSS) techniques
Rais and Viana (2011)	ChatGPT (GPT-4o)	A comprehensive survey of the applications of OR in healthcare
Tomasini and Van Wassenhove (2009)	ChatGPT (GPT-4o)	The document explores the critical role of SCM in humanitarian logistics
Alba et al. (2013)	ChatGPT (GPT-4o)	The document reviews recent advances and trends in parallel metaheuristics
Sinuany-Stern et al. (2000)	ChatGPT (GPT-4o)	The document introduces a hybrid AHP/DEA methodology for ranking decision-making units
Colorni et al. (1996)	ChatGPT (GPT-4o)	The document discusses the application of nature-inspired heuristics, such as GAs and simulated annealing, for solving complex combinatorial optimization problems
Gehring and Bortfeldt (1997)	ChatGPT (GPT-4o)	The document presents a GA designed to solve the container loading problem
Guajardo and Rönnqvist (2016)	ChatGPT (GPT-4o)	The document reviews various cost allocation methods in collaborative transportation
Osman and Christofides (1994)	ChatGPT (GPT-4o)	The document discusses solving capacitated clustering problems using a hybrid approach that combines simulated annealing and tabu search
Fagerholt (1999)	ChatGPT (GPT-4o)	The document presents an optimal fleet design method for a ship routing problem
Malaguti and Toth (2010)	ChatGPT (GPT-4o)	The document surveys algorithmic and computational results for the vertex coloring problem (VCP) and its generalizations
Archetti and Speranza (2012)	ChatGPT (GPT-4o)	The document surveys VRPs with split deliveries (SDVRP)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B1
(Continued)

Article	AI-ChatBot (model)	Topic extracted
Van Wassenhove and Pedraza Martinez (2012)	ChatGPT (GPT-4o)	The document explores how OR can adapt SCM best practices to humanitarian logistics
Rojas Viloria et al. (2021)	ChatGPT (GPT-4o)	The document provides a literature review on the use of unmanned aerial vehicles (UAVs; drones) in VRPs
Festa and Resende (2009a)	ChatGPT (GPT-4o)	The document provides an annotated bibliography of GRASP algorithms
Kozan and Preston (1999)	ChatGPT (GPT-4o)	The document applies GAs to optimize container transfer scheduling at multimodal terminals
Yu et al. (2018)	ChatGPT (GPT-4o)	The document extends the interactive and multicriteria decision-making (tomada de decisões interactiva e multicritério, TODIM) method to incorporate intuitionistic linguistic numbers (ILNs)
Narbón-Perpiñá and De Witte (2018a)	ChatGPT (GPT-4o)	The document systematically reviews literature on local government efficiency
Festa and Resende (2009b)	ChatGPT (GPT-4o)	The document provides an annotated bibliography of GRASP applications
Van der Vorst et al. (1998)	ChatGPT (GPT-4o)	The document investigates the impact of SCM on reducing uncertainties in food supply chains to improve logistical performance
de Freitas and Penna (2020)	ChatGPT (GPT-4o)	The document presents a variable neighborhood search algorithm for solving the flying sidekick traveling salesman problem (FSTSP)
Constantin and Florian (1995)	ChatGPT (GPT-4o)	The document presents a nonlinear bi-level programming approach for optimizing transit network frequencies
Yeh (2002)	ChatGPT (GPT-4o)	The document presents a problem-based approach for selecting MADM methods
Narbón-Perpiñá and De Witte (2018b)	ChatGPT (GPT-4o)	The document systematically reviews literature on local government efficiency
Mingers (2000)	ChatGPT (GPT-4o)	The document discusses the integration of soft and hard OR/MS methods
Vu et al. (2017)	ChatGPT (GPT-4o)	The document reviews surrogate-based methods for black-box optimization
Liberti et al. (2008)	ChatGPT (GPT-4o)	The document presents a branch-and-prune algorithm designed to solve the MDGP
Sörensen (2015)	CoPilot (GPT-4)	The document critiques the trend of using metaphors in metaheuristic research
Laporte et al. (2000)	CoPilot (GPT-4)	This document surveys classical and modern heuristics for the VRP, highlighting methods like savings algorithms, sweep algorithms, and tabu search heuristics
Bana e Costa and Vansnick (1994)	CoPilot (GPT-4)	MACBETH: An approach for constructing cardinal value functions through pairwise verbal judgments, simplifying the quantification of values and aiding decision-making
Chaudhry and Khan (2016)	CoPilot (GPT-4)	The document reviews FJSS techniques, covering various methods used since 1990 to address the problem
Rais and Viana (2011)	CoPilot (GPT-4)	This paper surveys applications of OR in healthcare

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B1
(Continued)

Article	AI-ChatBot (model)	Topic extracted
Tomasini and Van Wassenhove (2009)	CoPilot (GPT-4)	The main topic of this document is humanitarian logistics and its role in disaster relief
Alba et al. (2013)	CoPilot (GPT-4)	The main topic of this document is parallel metaheuristics
Sinuany-Stern et al. (2000)	CoPilot (GPT-4)	The paper presents a hybrid model combining DEA and AHP for ranking organizational units with multiple inputs and outputs
Colorni et al. (1996)	CoPilot (GPT-4)	Unable to analyze the file
Gehring and Bortfeldt (1997)	CoPilot (GPT-4)	Parallel GA for solving the container loading problem with a focus on strongly heterogeneous loads, demonstrating its effectiveness through comparative tests
Guajardo and Rönnqvist (2016)	CoPilot (GPT-4)	The main topic of this document is cost allocation methods in collaborative transportation
Osman and Christofides (1994)	CoPilot (GPT-4)	This document discusses the capacitated clustering problem (CCP) and introduces a hybrid simulated annealing and tabu search algorithm to solve it, highlighting its effectiveness through computational results
Fagerholt (1999)	CoPilot (GPT-4)	The main topic of this document is the problem of deciding an optimal fleet of ships in a liner shipping scenario, considering both fixed costs and operational costs
Malaguti and Toth (2010)	CoPilot (GPT-4)	The paper surveys algorithmic and computational results on the VCP and its generalizations
Archetti and Speranza (2012)	CoPilot (GPT-4)	The main topic of this document is the split delivery VRP (SDVRP)
Van Wassenhove and Pedraza Martinez (2012)	CoPilot (GPT-4)	The main topic of this document is the adaptation of SCM best practices to humanitarian logistics
Rojas Viloria et al. (2021)	CoPilot (GPT-4)	The main topic of this document is the analysis of UAVs routing problems
Festa and Resende (2009a)	CoPilot (GPT-4)	The main topic of this document is the GRASP, a metaheuristic for combinatorial optimization
Kozan and Preston (1999)	CoPilot (GPT-4)	The main topic of this document is optimizing container transfer schedules at multimodal terminals using GAs to reduce handling times and improve port efficiency
Yu et al. (2018)	CoPilot (GPT-4)	This document proposes an extended TODIM approach with ILNs for solving multicriteria decision-making (MCDM) problems
Narbón-Perpiñá and De Witte (2018a)	CoPilot (GPT-4)	The main topic of this document is the efficiency of local governments, covering studies from various countries and methods for measuring efficiency
Festa and Resende (2009b)	CoPilot (GPT-4)	This document provides an annotated bibliography of the GRASP literature
Van der Vorst et al. (1998)	CoPilot (GPT-4)	The main topic of this document is "Supply Chain Management in Food Chains: Improving Performance by Reducing Uncertainty"
de Freitas and Penna (2020)	CoPilot (GPT-4)	The document discusses the collaborative use of trucks and drones for parcel delivery, aiming to reduce delivery times and costs

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table B1
(Continued)

Article	AI-ChatBot (model)	Topic extracted
Constantin and Florian (1995)	CoPilot (GPT-4)	This document discusses the concept of data warehousing and its role in managing large volumes of data for decision-making purposes
Yeh (2002)	CoPilot (GPT-4)	This document discusses the topic of quality function deployment (QFD), a methodology for translating customer needs into product design requirements
Narbón-Perpiñá and De Witte (2018b)	CoPilot (GPT-4)	The main topic of this document is the efficiency of local governments, considering factors such as population density, income levels, tourism, and political ideology
Mingers (2000)	CoPilot (GPT-4)	The document discusses the integration of soft and hard OR/MS methods to address complex problems
Vu et al. (2017)	CoPilot (GPT-4)	The document surveys surrogate-based methods for black-box optimization
Liberti et al. (2008)	CoPilot (GPT-4)	A branch-and-prune algorithm for solving the MDGP

Appendix C

Reports generated by the AI-ChatBots

ChatGPT (GPT-4o)

This SLR analyzes the 30 most cited articles published in the journal **ITOR**. These articles cover a wide range of topics within the field of OR, with citations ranging from 731 to 109. The articles reviewed reveal key research areas and the evolution of methodologies in OR. The most cited article, with 731 citations, is by Sörensen (2015), which introduces novel metaheuristic methods, highlighting the importance of developing innovative optimization techniques in OR. Following this, Laporte et al. (2000) provide a comprehensive survey on heuristics applied to the vehicle routing problem (VRP), accumulating 518 citations. This work underscores the significance of heuristic approaches in solving complex routing problems, a recurrent theme in the literature. Bana e Costa and Vansnick (1994), with 327 citations, discuss techniques for the cardinal measurement of values, contributing to the theoretical foundation of decision analysis. Chaudhry and Khan (2016) also made a significant impact with 322 citations, offering a survey on the flexible job shop scheduling (FJSS) problem, reflecting the ongoing interest in optimizing scheduling processes in various industries. In the healthcare sector, Rais and Viana (2011) contribute with a survey on OR applications, amassing 274 citations. This work illustrates the growing intersection of OR and healthcare, particularly in improving efficiency and resource allocation. Tomasini and Van Wassenhove (2009), with 271 citations, focus on humanitarian logistics, addressing the challenges of logistics in crisis situations and the role of supply chain management (SCM) in humanitarian efforts. Surveys continue to play a crucial role in the field, as evidenced by Alba et al. (2013), who present a survey on parallel metaheuristics, garnering 245 citations. Similarly, Sinuany-Stern et al. (2000), with 241 citations, propose a combination of data envelopment analysis (DEA) and the analytic hierarchy process, showcasing the integration of multiple techniques to enhance

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

decision-making. Heuristic and evolutionary algorithms also feature prominently in this review. Colorni et al. (1996) explore nature-derived heuristics applied to graph optimization problems, which has been cited 218 times, while Gehring (1997) applies GAs to the container loading problem, receiving 209 citations. The versatility of GAs is further highlighted by Kozan (1999) in scheduling container problems, with 153 citations. A significant portion of the reviewed literature focuses on VRPs. Archetti and Speranza (2012) discuss vehicle routing with split deliveries, receiving 165 citations, while Rojas Viloria et al. (2021) explore the emerging area of drones in vehicle routing, cited 159 times. These works emphasize the ongoing innovation and relevance of routing problems in OR. SCM is another prominent theme. Van Wassenhove and Pedraza Martinez (2012) investigate humanitarian logistics, echoing the concerns of Tomasini and Van Wassenhove (2009), with both works collectively cited over 400 times. Van der Vorst et al. (1998) focus on logistical performance in food supply chains, contributing to the literature with 139 citations. In the realm of decision-making and optimization, Yu (2018) proposes an interactive multicriteria decision-making (MCDM) approach, garnering 150 citations, reflecting the importance of MCDM in complex decision scenarios. Similarly, Yeh (2002) explores the selection of compensatory multi-attribute decision-making (MADM) methods, cited 121 times. The contributions of Festa and Resende (2009a, 2009b), with 153 and 142 citations, respectively, provide extensive bibliographies on the greedy randomized adaptive search procedure (GRASP), furthering the development and application of metaheuristics. Several articles also delve into specific problem-solving methods. Liberti et al. (2008), with 109 citations, introduce the branch-and-prune algorithm for molecular distance geometry methods, showcasing applications beyond traditional OR domains. Vu et al. (2017), cited 109 times, explore surrogate-based methods for black-box optimization, reflecting the need for efficient optimization techniques in complex systems. Research on local government efficiency is represented by Narbón-Perpiñá and De Witte (2018a, 2018b), whose SLRs, with 149 and 116 citations, respectively, provide valuable insights into the efficiency of public sector operations. This focus on public sector applications demonstrates the broad applicability of OR methods. Last, discussions on the integration of soft methods in OR are presented by Mingers (2000), with 110 citations, highlighting the need for combining different methodological approaches. Constantin and Florian (1995), with 124 citations, contribute to the field with a nonlinear bi-level programming approach applied to transit networks, illustrating the complexity of real-world applications. In summary, the most cited articles in **ITOR** span a diverse range of topics, from metaheuristics and heuristic surveys to decision-making methods and SCM. These works collectively highlight the field's evolution, with a strong emphasis on the development of innovative algorithms and the application of OR methods to both traditional and emerging areas.

Claude ***3.5 Sonnet***

The SLR of the 30 most cited articles published in the scientific journal "*ITOR*" reveals several key insights. The review covers a wide range of topics, including novel metaheuristic methods, surveys on various operational research problems, techniques for cardinal measurement of values, proposals for combining DEA and analytic hierarchy process, the application of heuristics derived from nature to graph optimization problems, the use of GAs in container loading and scheduling problems, reviews on cost allocation methods and VRPs with split deliveries, the role of SCM in humanitarian logistics, literature reviews on the use of drones in VRPs, and systematic reviews

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table C1
Summary on the correctness of the report generated by the AI-ChatBots. X stands for missing citation, XX for error in the citation

Article	ChatGPT	Clade	Gemini	LeChat	CoPilot
Sörensen (2015)			X		
Laporte et al. (2000)			X		
Bana e Costa and Vansnick (1994)		XX	X	X	
Chaudhry and Khan (2016)			X		
Rais and Viana (2011)			X		
Tomasini and Van Wassenhove (2009)			X		
Alba et al. (2013)		X	X	X	
Sinuany-Stern et al. (2000)			X		
Colorni et al. (1996)		X	X		
Gehring and Bortfeldt (1997)			X		
Guajardo and Rönnqvist (2016)	X	X	X	X	
Osman and Christofides (1994)	X	X	X	X	
Fagerholt (1999)	X	X	X	X	
Malaguti and Toth (2010)	X		X	X	
Archetti and Speranza (2012)		X	X	X	
Van Wassenhove and Pedraza Martinez (2012)			X	X	
Rojas Viloria et al. (2021)			X	X	
Festa and Resende (2009a)			X		
Kozan and Preston (1999)			X		
Yu et al. (2018)			X		
Narbón-Perpiñá and De Witte (2018a)			X	X	
Festa and Resende (2009b)			X		
Van der Vorst et al. (1998)			X	X	
de Freitas and Penna (2020)	X	X	X		
Constantin and Florian (1995)		X	X	X	
Yeh (2002)		X	X	X	
Narbón-Perpiñá and De Witte (2018b)			X	X	
Mingers (2000)			X	X	
Vu et al. (2017)			X	X	
Liberti et al. (2008)		X	X	X	

on local governments' efficiency. The article with the highest number of citations is Sörensen (2015), which discusses novel metaheuristic methods. This is followed by Laporte et al. (2000), which provides a survey on heuristics for the VRP, and Bana e Costa and Vansnick (1994), which presents techniques for cardinal measurement of values. Several articles focus on surveying different operational research problems, such as the FJSS problem (Chaudhry and Khan, 2016), OR in healthcare (Rais and Viana, 2011), and vertex coloring problems (VCPs) (Malaguti and Toth, 2010). The review also includes articles that propose novel approaches, such as the combination of DEA and analytic hierarchy process (Sinuany-Stern et al., 2000), the application of GAs to container loading and scheduling problems (Gehring, 1997; Kozan, 1999), and an interactive MCDM approach (Yu, 2018). The review also covers various aspects of SCM, including humanitarian logistics (Tomasini and Van Wassenhove, 2009; Van Wassenhove and Pedraza Martinez, 2012) and

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

logistical performance in food supply chains (Van der Vorst et al., 1998). Additionally, there are articles that focus on SLRs, such as the ones on drones in VRPs (Rojas Viloria et al., 2021) and local governments' efficiency (Narbón-Perpiñá and De Witte, 2018a, 2018b). The review also includes articles that discuss methodological aspects, such as the bibliography and survey of GRASP (Festa and Resende, 2009a, 2009b), the use of surrogate-based methods for black-box optimization (Vu et al., 2017), and the discussion on soft methods and their combinations (Mingers, 2000). Overall, the SLR highlights the diverse range of topics and approaches covered in the field of operational research as reflected in the highly cited articles published in the "*ITOR*" journal.

Gemini (1.5 Flash)

Analysis of top-cited articles in ITOR. An SLR of the 30 most cited articles published in *ITOR* reveals a strong focus on specific areas within the field of OR.

Metaheuristics and optimization. A significant number of articles explore various metaheuristic methods, including GAs, simulated annealing, and tabu search, for solving complex optimization problems. These methods are frequently applied to VRPs, scheduling problems, and container loading problems.

Surveys and reviews. Several papers provide comprehensive surveys on specific topics within OR, such as heuristic methods for VRPs, FJSS problems, and cost allocation methods. These surveys serve as valuable resources for researchers and practitioners seeking an overview of the state-of-the-art in these areas.

Humanitarian logistics. A growing body of research focuses on humanitarian logistics, addressing challenges such as disaster relief and aid distribution. The articles explore SCM, VRPs, and decision-making processes in humanitarian contexts.

Other notable topics. Other topics covered in the most cited articles include DEA, MCDM, and mathematical programming techniques. These articles contribute to the development of tools and methodologies for addressing a wide range of operational challenges.

Overall trends. In conclusion, the analysis of the top-cited articles in *ITOR* highlights a strong emphasis on metaheuristics, optimization, and specific application areas such as vehicle routing and scheduling. The growing interest in humanitarian logistics reflects the increasing importance of addressing societal challenges through OR.

LeChat (Large 2). In this SLR, we analyzed the 30 most cited articles published in the scientific journal "*ITOR*." The articles covered a wide range of topics, with a notable focus on surveys, metaheuristics, and specific applications of operational research methods. The most cited article, with 731 citations, was Sörensen (2015), which discussed "novel" metaheuristic methods, indicating

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

a strong interest in innovative optimization techniques. Several other highly cited articles were surveys, including Laporte et al. (2000) with 518 citations on heuristics for the VRP, and Chaudhry and Khan (2016) with 322 citations on the FJSS problem. These surveys provide comprehensive overviews of their respective fields, highlighting their significance in the academic community. Articles focusing on specific applications also received substantial attention. For instance, Rais and Viana (2011) with 274 citations, reviewed OR in healthcare, while Tomasini and Van Wassenhove (2009) with 271 citations, discussed humanitarian logistics. These articles demonstrate the practical relevance of operational research in addressing real-world problems. Additionally, articles proposing new methods or combinations of existing methods were well-cited. Sinuany-Stern et al. (2000) with 241 citations, proposed a combination of data envelopment analysis (DEA) and the analytic hierarchy process (AHP), and Yu (2018) with 150 citations, introduced an interactive multi-criteria decision-making (MCDM) approach. Some articles focused on nature-inspired heuristics, such as Colorni et al. (1996) with 218 citations, which applied heuristics derived from nature to graph optimization problems. GAs were also a recurring theme, with Gehring (1997) and Kozan (1999) applying them to container loading and scheduling problems, respectively. Reviews and bibliographies on specific methods like GRASP, by Festa and Resende (2009a, 2009b), were also highly cited, indicating their importance in the field.

In summary, the most cited articles in "*ITOR*" reflect a strong interest in surveys, metaheuristics, and practical applications of operational research. The diversity of topics and methods highlights the journal's broad scope and influence in the operational research community.

CoPilot (GPT-4). The SLR of the 30 most cited articles published in the "*ITOR*" reveals a diverse range of topics and methodologies within the field. The most cited article by Sörensen (2015) with 731 citations focuses on novel metaheuristic methods, highlighting the importance and innovation in this area. Laporte et al. (2000) with 518 citations provide a comprehensive survey on heuristics for the VRP, indicating the ongoing relevance of this classic optimization challenge. Bana e Costa and Vansnick (1994) with 327 citations discuss techniques for cardinal measurement of values, contributing significantly to decision analysis. Chaudhry and Khan (2016) with 322 citations offer a survey on FJSS problems, reflecting the complexity and practical importance of scheduling in OR. Rais and Viana (2011) with 274 citations review OR applications in healthcare, showcasing the critical role of OR in improving healthcare systems. Tomasini and Van Wassenhove (2009) with 271 citations delve into humanitarian logistics, emphasizing the application of OR in disaster relief and humanitarian efforts. Alba et al. (2013) with 245 citations survey parallel metaheuristics, underscoring advancements in computational methods. Sinuany-Stern et al. (2000) with 241 citations propose a combination of DEA and analytic hierarchy process, contributing to efficiency analysis. Colorni et al. (1996) with 218 citations explore nature-inspired heuristics for graph optimization problems, reflecting the interdisciplinary nature of OR. Gehring (1997) with 209 citations applies GAs to the container loading problem, demonstrating the practical applications of evolutionary algorithms. Guajardo and Rönnqvist (2016) with 204 citations review cost allocation methods, highlighting the importance of fair and efficient resource distribution. Osman and Christofides (1994) with 188 citations discuss the capacitated clustering problem, contributing to clustering and partitioning methodologies. Fagerholt (1999) with 178 citations addresses optimal fleet design in ship routing, reflecting the significance of maritime logistics. Malaguti and Toth (2010) with 177

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

citations survey VCPs, contributing to graph theory and combinatorial optimization. Archetti and Speranza (2012) with 165 citations focus on VRPs with split deliveries, addressing practical logistics challenges. Van Wassenhove and Pedraza Martinez (2012) with 165 citations discuss SCM in humanitarian logistics, further emphasizing the role of OR in humanitarian efforts. Rojas Viloria et al. (2021) with 159 citations review the use of drones in VRPs, showcasing the integration of new technologies in logistics. Festa and Resende (2009a) with 153 citations provide a bibliography and survey of GRASP, contributing to metaheuristic research. Kozan (1999) with 153 citations applies GAs to scheduling container problems, reflecting the practical applications of GAs in logistics. Yu (2018) with 150 citations proposes an interactive MCDM approach, contributing to decision support systems. Narbón-Perpiñá and De Witte (2018a) with 149 citations conducts an SLR on local governments' efficiency, highlighting the application of OR in public administration. Festa and Resende (2009b) with 142 citations continue their survey of GRASP, further contributing to the understanding of this metaheuristic. Van der Vorst et al. (1998) with 139 citations discuss SCM and logistical performance in food supply chains, reflecting the importance of OR in food logistics. De Freitas et al. (2020) with 137 citations apply variable neighborhood search (VNS) to the flying sidekick traveling salesman problem (FSTSP), showcasing advancements in combinatorial optimization. Constantin and Florian (1995) with 124 citations propose a nonlinear bi-level programming approach for transit networks, contributing to transportation planning. Yeh (2002) with 121 citations discusses the selection of compensatory MADM methods, contributing to decision analysis. Narbón-Perpiñá and De Witte (2018b) with 116 citations continue their SLR on local governments' efficiency, further emphasizing the role of OR in public administration. Mingers (2000) with 110 citations discusses soft methods and their combinations, contributing to the understanding of qualitative methods in OR. Vu et al. (2017) with 109 citations explore surrogate-based methods for black-box optimization, reflecting advancements in optimization techniques. Liberti et al. (2008) with 109 citations propose a branch-and-prune algorithm for molecular distance geometry methods, showcasing the interdisciplinary applications of OR in molecular biology. This review highlights the breadth and depth of research published in the journal, covering a wide range of topics from metaheuristics and scheduling to healthcare and humanitarian logistics, demonstrating the diverse applications and ongoing advancements in the field of OR.

Appendix D

Report generated by the Gemini 2.0 experimental (generated on January 16, 2025)

This report presents an SLR of the 30 most cited articles published in *ITOR*. The analysis reveals several prominent themes within the field. Methodologically, metaheuristics are a dominant area of research, with Sörensen (2015) leading with 731 citations, focusing on "novel" metaheuristic methods. Parallel metaheuristics (Alba et al., 2013) and nature-inspired heuristics applied to graph optimization problems (Colorni et al., 1996) also feature prominently. Specific metaheuristic applications include GAs applied to container loading (Gehring and Bortfeldt, 1997; Kozan and Preston, 1999) and VNS for the FSTSP (de Freitas and Penna, 2020). The GRASP is extensively covered in two parts by Festa and Resende (2009a, 2009b). Surveys and reviews are a significant portion of the highly cited articles. Laporte et al. (2000) offer a highly cited survey on heuristics for

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

VRPs, while Chaudhry and Khan (2016) survey FJSS. Rais and Viana (2011) review OR in healthcare. Guajardo and Rönnqvist (2016) provide a review on cost allocation methods, and Malaguti and Toth (2010) present a survey on VCPs. VRPs are a recurring theme, with contributions on split deliveries (Archetti and Speranza, 2012) and the integration of drones (Rojas Viloria et al., 2021). The capacitated clustering problem is discussed by Osman and Christofides (1994). Humanitarian logistics is another important area, with contributions from Tomasini and Van Wassenhove (2009) and Van Wassenhove and Pedraza Martinez (2012). SCM is also represented, with a focus on logistical performance in food supply chains (Van der Vorst et al., 1998). MCDM is addressed by Bana e Costa and Vansnick (1994) on cardinal measurement of values, Sinuany-Stern et al. (2000) combining DEA and the Analytic Hierarchy Process, Yu et al. (2018) proposing an interactive MCDM approach, and Yeh (2002) discussing the selection of compensatory MADM methods. Other topics include optimal fleet design in ship routing (Fagerholt, 1999), nonlinear bi-level programming applied to transit networks (Constantin and Florian, 1995), SLRs on local governments' efficiency (Narbón-Perpiñá and De Witte, 2018a, 2018b), discussions on soft methods and their combinations (Mingers, 2000), surrogate-based methods for black-box optimization (Vu et al., 2017), and a branch-and-prune algorithm for molecular distance geometry methods (Liberti et al., 2008).

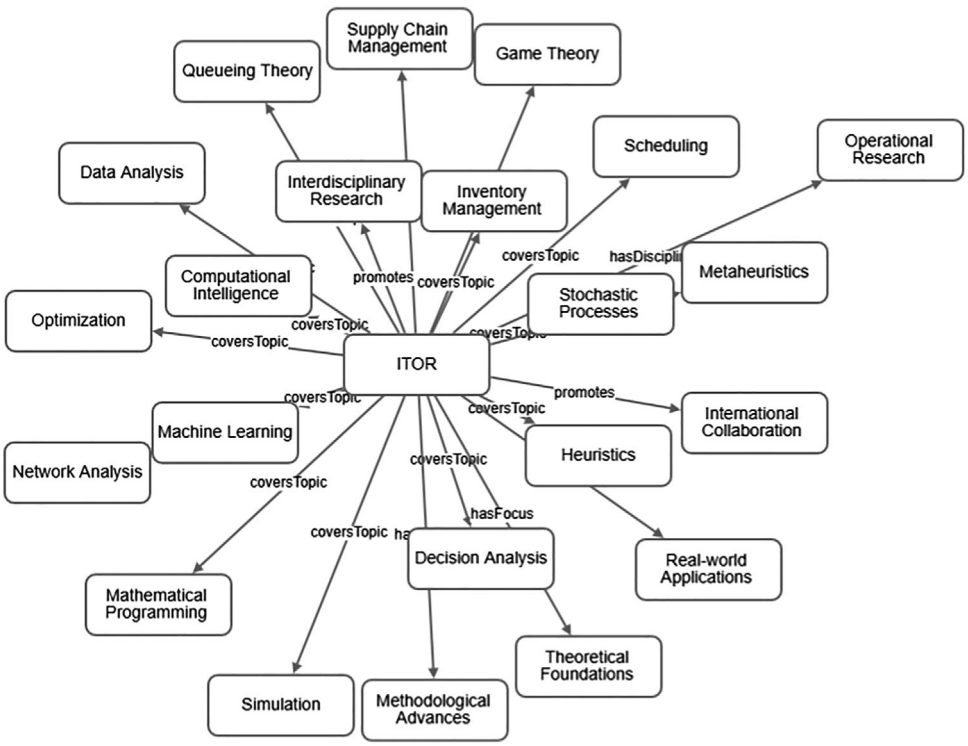

Fig. D1. Conceptual map of *ITOR* generated by Gemini (Flash 2.0 experimental). The map was generated on January 16, 2025.

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table D1
Results of the data extraction phase using LeChat. Date of execution January 16, 2024

Reference	Topic extracted by LeChat
Sörensen (2015)	The paper criticizes the trend of creating "novel" metaheuristic methods based on metaphors, arguing that they often lack genuine innovation and scientific rigor, and calls for more critical evaluation in the field
Laporte et al. (2000)	The paper reviews classical and modern heuristics for the VRP, highlighting the evolution from traditional methods like the savings algorithm and sweep algorithm to advanced metaheuristics
Bana e Costa and Vansnick (1994)	The paper introduces MACBETH, an interactive approach for constructing cardinal value functions through verbal judgments of attractiveness differences
Chaudhry and Khan (2016)	The paper reviews the extensive research on flexible job shop scheduling (FJSSP) over the past 25 years, highlighting various techniques from exact methods to hybrid approaches
Rais and Viana (2011)	This paper surveys the application of OR in healthcare, focusing on optimization issues such as resource allocation, strategic planning, and operational aspects like scheduling and treatment planning
Tomasini and Van Wassenhove (2009)	This paper discusses the evolution of SCM in disaster relief, highlighting the role of new players like the private sector in improving efficiency and effectiveness
Alba et al. (2013)	This paper reviews recent advances and new trends in parallel metaheuristics, focusing on how new technologies and parallel computing platforms are being utilized to improve optimization algorithms
Sinuany-Stern et al. (2000)	This paper presents a methodology that combines DEA and the AHP to fully rank organizational units based on multiple inputs and outputs
Colorni et al. (1996)	This paper explores heuristic algorithms inspired by nature, such as GAs and simulated annealing, for solving complex combinatorial optimization problems
Gehring and Bortfeldt (1997)	This paper presents a GA designed to solve the container loading problem, focusing on arranging heterogeneous boxes into a single container while considering practical constraints
Guajardo and Rönnqvist (2016)	This paper reviews various cost allocation methods used in collaborative transportation, focusing on how costs or benefits are shared among participants in different logistics scenarios
Osman and Christofides (1994)	This paper introduces a hybrid simulated annealing and tabu search algorithm to solve the capacitated clustering problem
Fagerholt (1999)	The paper presents a method for optimizing the fleet design in a liner shipping problem, focusing on determining the optimal types and number of ships and their weekly routes to minimize costs
Malaguti and Toth (2010)	The paper surveys key algorithmic and computational results on the VCP and its generalizations
Archetti and Speranza (2012)	The paper surveys the VRP with SDVRP, where each customer can be served by multiple vehicles
Van Wassenhove and Pedraza Martinez (2012)	The paper discusses how OR can be used to adapt SCM best practices to humanitarian logistics, which faces unique challenges such as demand and supply uncertainty, high decentralization, and multiple parties with different objectives
Rojas Viloria et al. (2021)	This document is a literature review on the use of UAVs, or drones, in VRPs

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table D1
(Continued)

Reference	Topic extracted by LeChat
Festa and Resende (2009a)	This document is an annotated bibliography focusing on the algorithms and enhancements of the GRASP, a metaheuristic for combinatorial optimization
Kozan and Preston (1999)	This document presents a GA approach to optimize container transfer scheduling at multimodal terminals, aiming to reduce handling times and ship berthing durations
Yu et al. (2018)	The paper proposes an extended TODIM approach using ILNs and nonlinear programming for MCDM problems
Narbón-Perpiñá and De Witte (2018a)	This paper provides a comprehensive review of studies on local government efficiency from 1990 to 2016, focusing on measurement techniques and input-output variables
Festa and Resende (2009b)	This document is an annotated bibliography focusing on the applications of the GRASP, a metaheuristic for combinatorial optimization
Van der Vorst et al. (1998)	This paper explores how SCM can improve logistical performance in food supply chains by reducing uncertainties
de freitas and penna (2020)	this paper introduces a hybrid heuristic for the FSTSP, which combines a truck and a drone for parcel delivery
Constantin and Florian (1995)	The paper presents a method for optimizing transit line frequencies in urban networks using a nonlinear bi-level programming approach, aiming to minimize total travel and waiting times while considering fleet size constraints
Yeh (2002)	The paper introduces a method for selecting the most appropriate MADM method for a specific problem
Narbón-Perpiñá and De Witte (2018b)	The paper reviews the literature on local governments' efficiency, focusing on the methods used to measure efficiency and the impact of environmental variables
Mingers (2000)	The paper discusses the integration of soft (qualitative) and hard (quantitative) OR/MS methods to tackle complex, real-world problems effectively
Vu et al. (2017)	The paper discusses surrogate-based methods for black-box optimization, focusing on iteratively constructing surrogate models to approximate expensive-to-evaluate functions
Liberti et al. (2008)	The paper introduces a branch-and-prune algorithm for solving the MDGP, which involves determining the 3D positions of atoms in a molecule given some inter-atomic distances

Table D2
Queries obtained by the different Gemini AI version for Scopus and WOS search.

ChatBot (model)	WOS	Scopus
Gemini (2.0 flash-experimental)- executed on 02/08/2024	SO = ("International Transactions in Operational Research")	SRCTITLE("International Transactions in Operational Research")
Gemini (1.5 flash) - executed on 02/08/2024	TS = (International Transactions in Operational Research) AND DOCUMENT TYPE(Article)	TITLE("International Transactions in Operational Research") AND DOCTYPE(ar)
Gemini (1.5 flash) - executed on 16/01/2025	TS = (International Transactions in Operational Research) AND (Times Cited>0)	TITLE-ABS-KEY("International Transactions in Operational Research") AND (CITED>0)

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix E

TableE2

Table E1
Prompts used for sensitivity analysis in Scenario 2 of the search phase by the various chatbots

Prompt number	Prompt
1	*You are the editor-in-chief of the journal "International Transactions in Operational Research". You are searching for the 30 most important articles in your journal based on the number of citations. Provide me with a query to use in Web of Science or Scopus to retrieve these articles*
2	*You are the editor-in-chief of the journal "International Transactions in Operational Research". You are searching for the 30 most important articles in your journal based on the number of citations. Provide me with the queries to use in Web of Science or Scopus to retrieve these articles*
3	*Provide me with the queries to use in Web of Science or Scopus to retrieve the 30 most cited article from the journal "International Transactions in Operational Research"*
4	*You are the editor-in-chief of the journal "International Transactions in Operational Research". You are searching for the 30 most important articles in your journal based on the number of citations. What will you do? If you need to use some queries, please provide them*
5	*You are the editor-in-chief of the journal "International Transactions in Operational Research". You are searching for the 30 most important articles in your journal based on the number of citations. Provide me two queries to retrieve these articles, one for the Web of Science and another one for Scopus.*

Table E2
Results of the query generation for the search phase (Scenario 2) using the prompts given in Table E1. Date of execution January 16, 2025. In the rows where *"1 (executed on August 2, 2024)"* is shown in the prompt column, the queries are the same as those shown in Table 3.

Prompt	WOS	Scopus	ChatBot (Model)
1	SO = ("International Transactions in Operational Research")	SOURCE-TITLE("International Transactions in Operational Research")	ChatGPT (GPT-4o)
2	SO = ("International Transactions in Operational Research")	SOURCE("International Transactions in Operational Research")	ChatGPT (GPT-4o)
3	SO = ("International Transactions in Operational Research")	SOURCE-TITLE("International Transactions in Operational Research")	ChatGPT (GPT-4o)
4	SO = ("International Transactions in Operational Research") AND DOCUMENT TYPES: (Article) Sort by: Times Cited – Highest to Lowest	SOURCE-TITLE("International Transactions in Operational Research") AND DOCTYPE(ar) Sort by: Cited by (highest)	ChatGPT (GPT-4o)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table E2
(Continued)

Prompt	WOS	Scopus	ChatBot (Model)
5	Publication Name: "International Transactions in Operational Research" Sort by: Times Cited (highest to lowest) Retrieve: Top 30 results	SOURCE-TITLE("International Transactions in Operational Research") SORT: Cited by (highest) LIMIT: 30	ChatGPT (GPT-4o)
1 (executed on August 2, 2024)	SO = ("International Transactions in Operational Research") SORT BY: Times Cited – Highest to Lowest	SOURCE-TITLE("International Transactions in Operational Research") SORT BY: Cited by (highest)	ChatGPT (GPT-4o)
1	SO = ("International Transactions in Operational Research")	SRCTITLE("International Transactions in Operational Research")	Claude (3.5 Sonnet)
2	SO = ("International Transactions in Operational Research") Sort by: Times Cited - highest to lowest	SRCTITLE("International Transactions in Operational Research") Sort by: Cited by (highest)	Claude (3.5 Sonnet)
3	SO = ("International Transactions in Operational Research")	SRCTITLE("International Transactions in Operational Research")	Claude (3.5 Sonnet)
4	SO = ("International Transactions in Operational Research") OR SO = ("ITOR")	SRCTITLE("International Transactions in Operational Research") OR SRCTITLE("ITOR")	Claude (3.5 Sonnet)
5	SO = ("International Transactions in Operational Research") Sort by: Times Cited - highest to lowest	SRCTITLE("International Transactions in Operational Research") Sort by: Cited by (highest)	Claude (3.5 Sonnet)
1-old (executed on August 2, 2024	SO = "International Transactions in Operational Research"	SRCTITLE("International Transactions in Operational Research")	Claude (3.5 Sonnet)
1	TS = ("International Transactions in Operational Research") AND PY = 1994-2024	TITLE-ABS-KEY("International Transactions in Operational Research") AND PUBYEAR>1993 AND PUBYEAR<2025	Copilot (GPT-4)
2	TS = ("International Transactions in Operational Research") AND PY = 2000-2025	SOURCE("International Transactions in Operational Research") AND PUBYEAR>1999 AND PUBYEAR<2025	Copilot (GPT-4)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table E2
(Continued)

Prompt	WOS	Scopus	ChatBot (Model)
3	TS = ("International Transactions in Operational Research") AND PY = 2000-2025	SOURCE = ("International Transactions in Operational Research") AND PUBYEAR>1999 AND PUBYEAR<2025	Copilot (GPT-4)
4	TS = ("International Transactions in Operational Research") AND PY = *	TITLE-ABS-KEY("International Transactions in Operational Research") AND PUBYEAR IS NOT NULL	Copilot (GPT-4)
5	TS = ("International Transactions in Operational Research") AND PY = 2000-2025	SOURCE = ("International Transactions in Operational Research") AND PUBYEAR>1999 AND PUBYEAR<2026	Copilot (GPT-4)
1-old (executed on August 2, 2024	SO = ("International Transactions in Operational Research") AND PY = (1994-2024)	SOURCE-ID(9700153238) AND PUBYEAR > 1993	Copilot (GPT-4)
1	TS = (International Transactions in Operational Research) AND (Times Cited>0)	TITLE-ABS-KEY("International Transactions in Operational Research") AND (CITED>0)	Gemini (1.5 flash)
2	SO = "International Transactions in Operational Research"	SRCTITLE("International Transactions in Operational Research")	Gemini (1.5 flash)
3	SO = International Transactions in Operational Research	SRCTITLE(International Transactions in Operational Research)	Gemini (1.5 flash)
4	SO = "International Transactions in Operational Research"	SRCTITLE("International Transactions in Operational Research")	Gemini (1.5 flash)
5	SO = ("International Transactions in Operational Research") AND DOCUMENT TYPES: (Article)	SRCTITLE ("International Transactions in Operational Research") AND (DOCTYPE (ar))	Gemini (1.5 flash)
1-old (executed on August 2, 2024	TS = (International Transactions in Operational Research) AND DOCUMENT TYPE(Article)	TITLE("International Transactions in Operational Research") AND DOCTYPE(ar)	Gemini (1.5 flash)
1	SO = ("International Transactions in Operational Research")	SRCTITLE("International Transactions in Operational Research")	Gemini (2.0 flash-experimental)

Continued

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table E2
(Continued)

Prompt	WOS	Scopus	ChatBot (Model)
2	SO = "International Transactions in Operational Research"	SRCTITLE("International Transactions in Operational Research")	Gemini (2.0 flash-experimental)
3	SO = "International Transactions in Operational Research"	SRCTITLE("International Transactions in Operational Research")	Gemini (2.0 flash-experimental)
4	Publication Name: "International Transactions in Operational Research"	TITLE-ABS-KEY ("International Transactions in Operational Research")	Gemini (2.0 flash-experimental)
5	SO = ("International Transactions in Operational Research") AND PY = (2000-2024) SORT BY Citation Count DESC	SRCTITLE ("International Transactions in Operational Research") AND (PUBYEAR > 1999 AND PUBYEAR < 2025) SORT BY Cited by-Count	Gemini (2.0 flash-experimental)
1	SO = "INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH"	SRCTITLE("International Transactions in Operational Research")	LeChat (Large 2)
2	SO = "INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH"	SOURCE-ID(21100207338)	LeChat (Large 2)
3	SO = INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH	SRCTITLE("International Transactions in Operational Research")	LeChat (Large 2)
4	SO = "INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH" AND DT = (ARTICLE)	SRCTITLE("International Transactions in Operational Research") AND DOCTYPE(ar)	LeChat (Large 2)
5	SO = "INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH" AND DT = (Article)	SRCTITLE("International Transactions in Operational Research") AND DOCTYPE(ar)	LeChat (Large 2)
1-old (executed on August 2, 2024	SO = (International Transactions in Operational Research)	SOURCE-ID("International Transactions in Operational Research")	LeChat (Large 2)

© 2025 The Author(s).
International Transactions in Operational Research published by John Wiley & Sons Ltd on behalf of International Federation of Operational Research Societies.

14753995, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/itor.70054 by Cochrane Taiwan, Wiley Online Library on [03/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License